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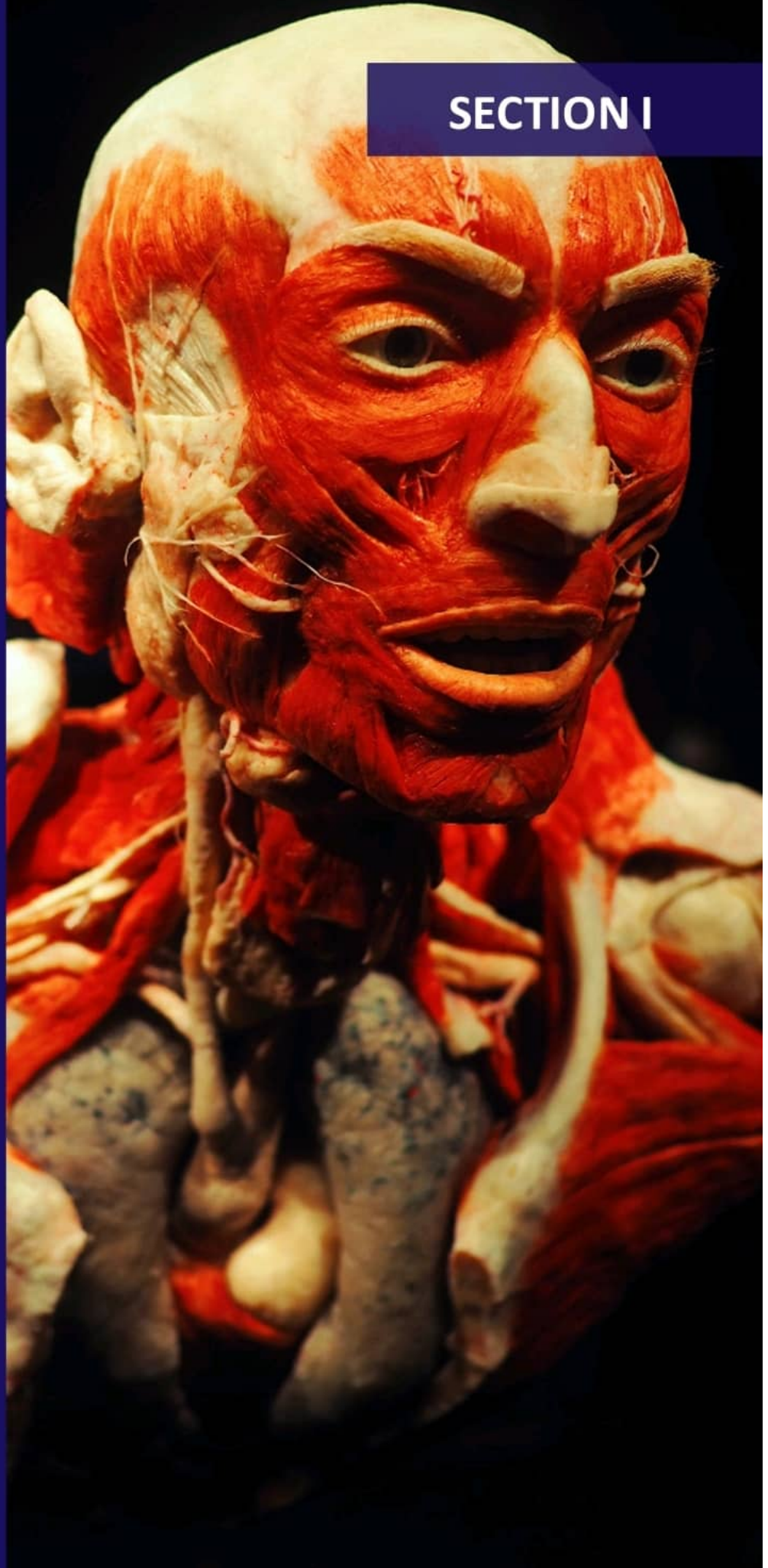
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ANATOMY SECTION

SECTION I





THORACIC CAGE

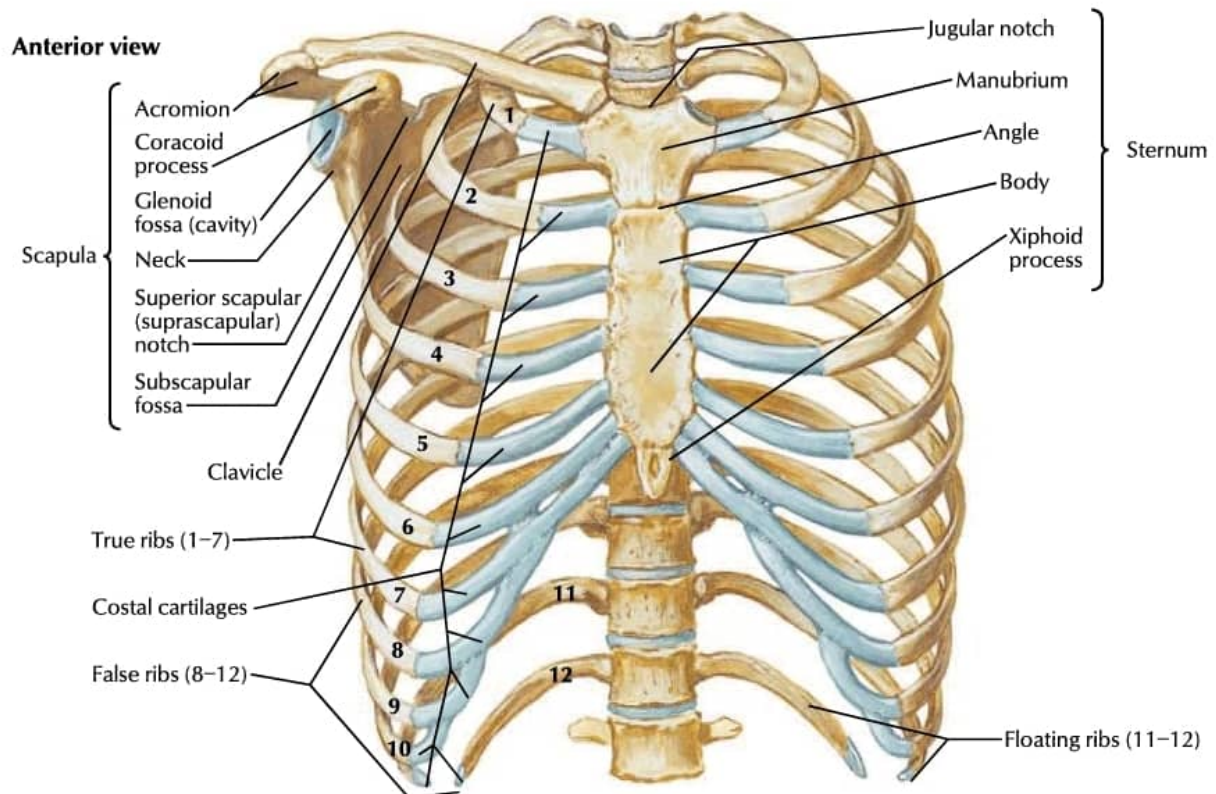
- The skeleton of the thorax forms **an osteocartilaginous framework** which is conical in shape with its anterior wall shorter than its posterior wall

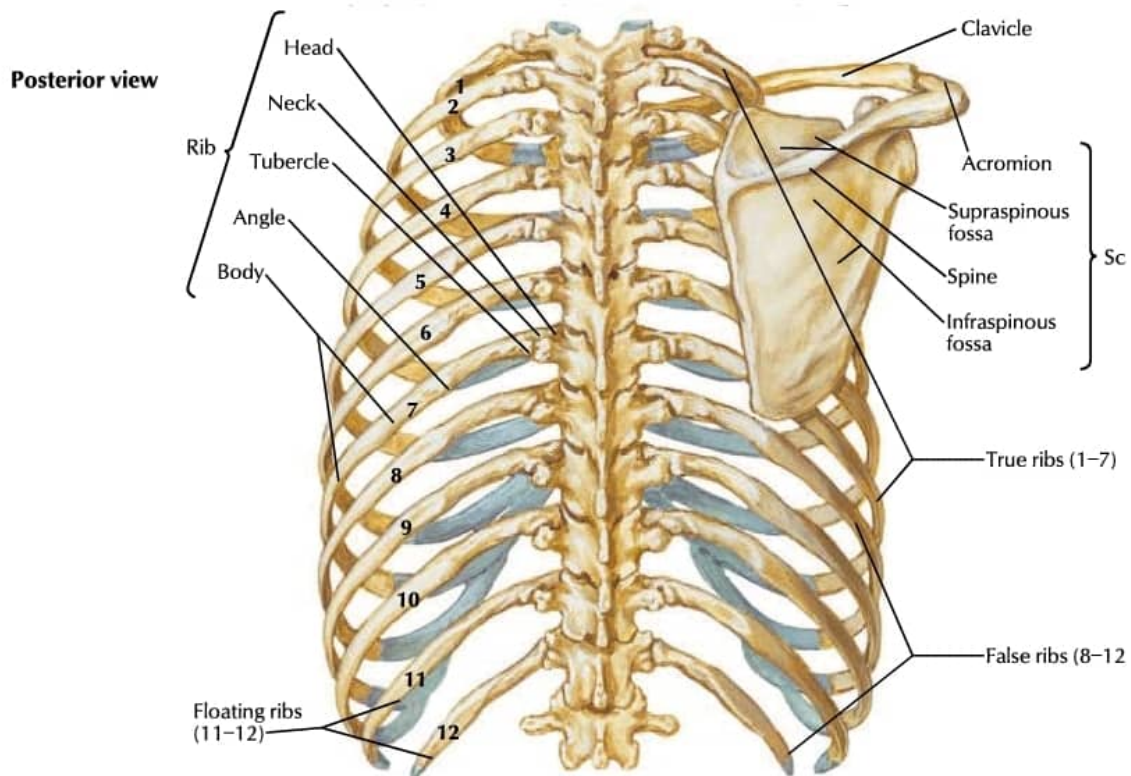
- Formation of Thoracic cage:

- **Anteriorly:** Sternum and costal cartilages
- **Posteriorly:** Thoracic vertebrae, intervertebral discs and ribs
- **Laterally:** Twelve pairs of ribs, mainly the shaft

- Function of Thoracic cage:

1. **Supports and protects** thoracic and upper abdominal viscera.
2. Shares in **movements of respiration.**





PARTS OF THE THORACIC CAVITY

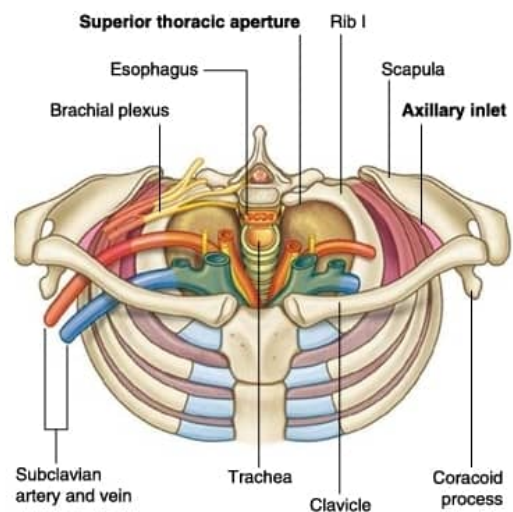
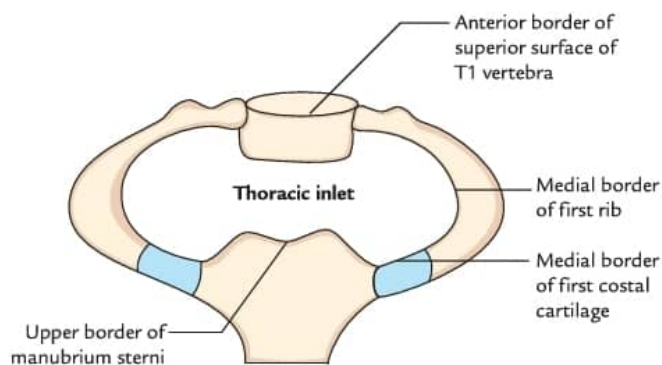
- 1. An inlet:** **Narrow**, continuous with the root of the neck.
- 2. A cavity:** Encloses the constituents of the thorax e.g. lungs and heart.
- 3. An outlet:** **Broad**, closed by diaphragm which separates it from abdominal cavity.

THORACIC INLET

- It is the upper, narrow opening connecting the thoracic cavity **with the root of the neck**.
- **Its plane** is oblique, sloping downwards and forwards

Boundaries

- **Anteriorly:** Upper border of manubrium sterni.
- **Posteriorly:** Body of 1st thoracic vertebra.
- **Laterally:** Inner border of 1st rib.



Structures passing through the inlet

I- In the middle of the inlet:

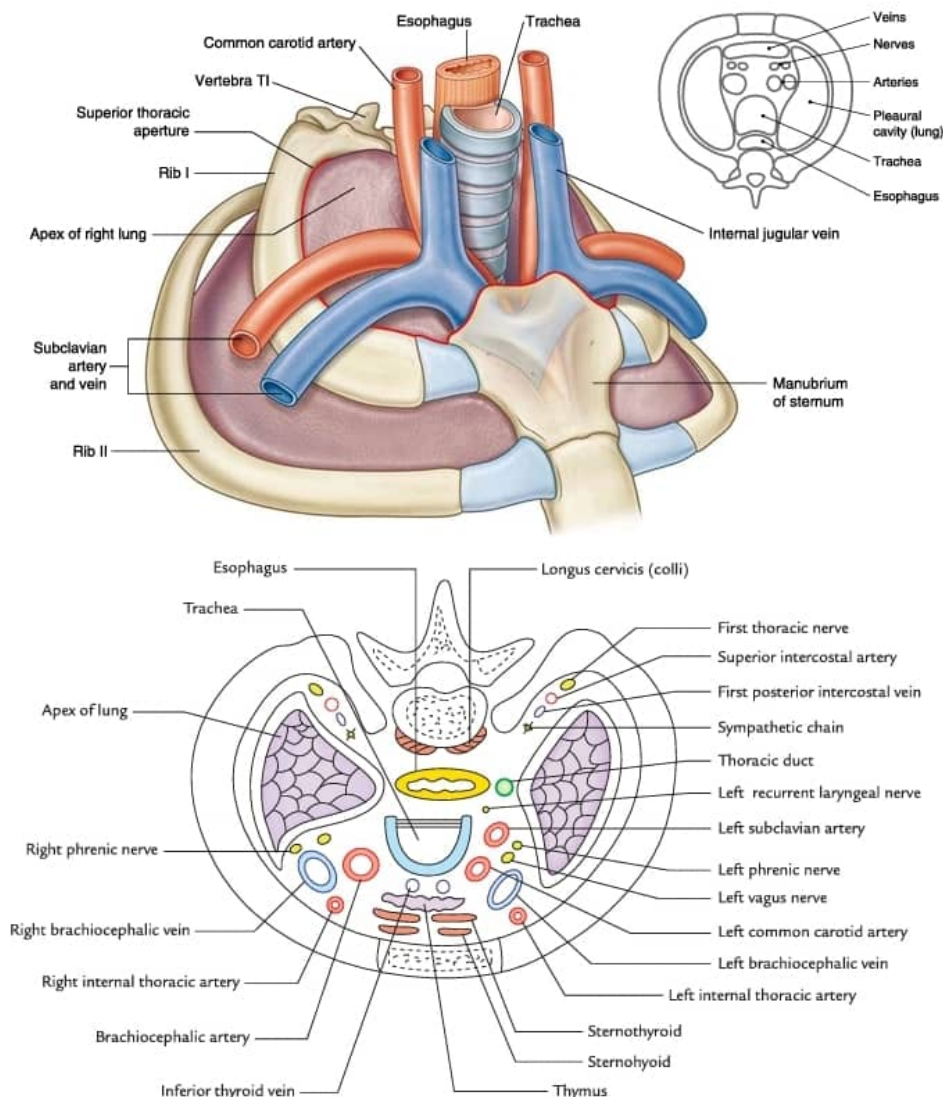
1. Trachea and oesophagus with Left recurrent laryngeal nerve in between.
2. Thoracic duct: **behind left margin of oesophagus.**

II- On each side, most laterally:

1. **Anteriorly:** Apex of the lung and Internal thoracic (internal mammary) artery.
2. Three structures behind the lung, in front of the neck of the 1st rib:
 - a. **Medially:** Sympathetic chain.
 - b. **Laterally:** Ventral 1st ramus of T1.
 - c. **In between:** Superior intercostal artery.

III- Between lungs and midline structures:

1. **Vessels:**
 - a. **On the right:** Brachiocephalic artery.
 - b. **On the left:** Left common carotid and left subclavian artery.
 - c. **On either side:** Left and right brachiocephalic veins.
2. **Nerves:** **On either side:** Left and right phrenic and vagi nerves.



Structures passing through the inlet



Suprapleural Membrane (Sibson's Fascia)

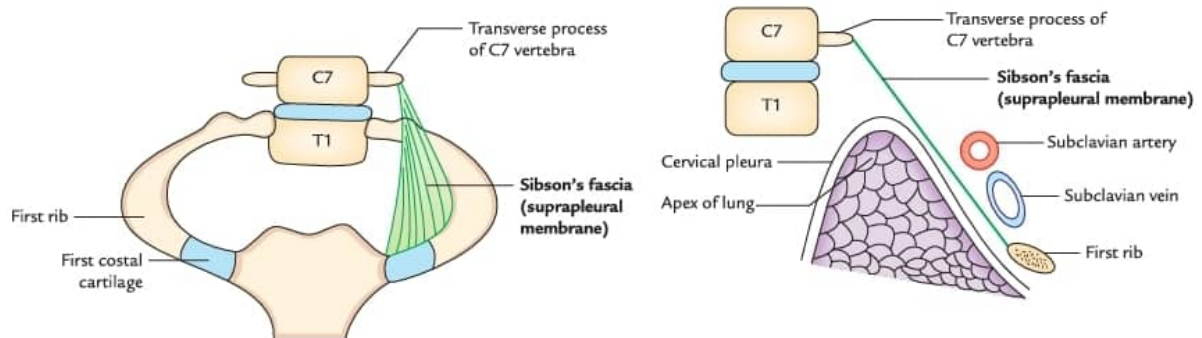
- **Definition:** Triangular in shape having an apex (**posteriorly**) and a base (**anterolaterally**).

- **Attachments:**

- The **apex**: is attached to the transverse process of **C7 vertebrae**.
- The **base**: is attached to the inner border of the **1st rib and its costal cartilage**.

- **Functions:**

- Protects** the cervical pleura and apex of the lung.
- Prevents inward suction** of structures at root of neck during inspiration.
- Prevents upward bulging** of apex of lung during forcible expulsive expiration.



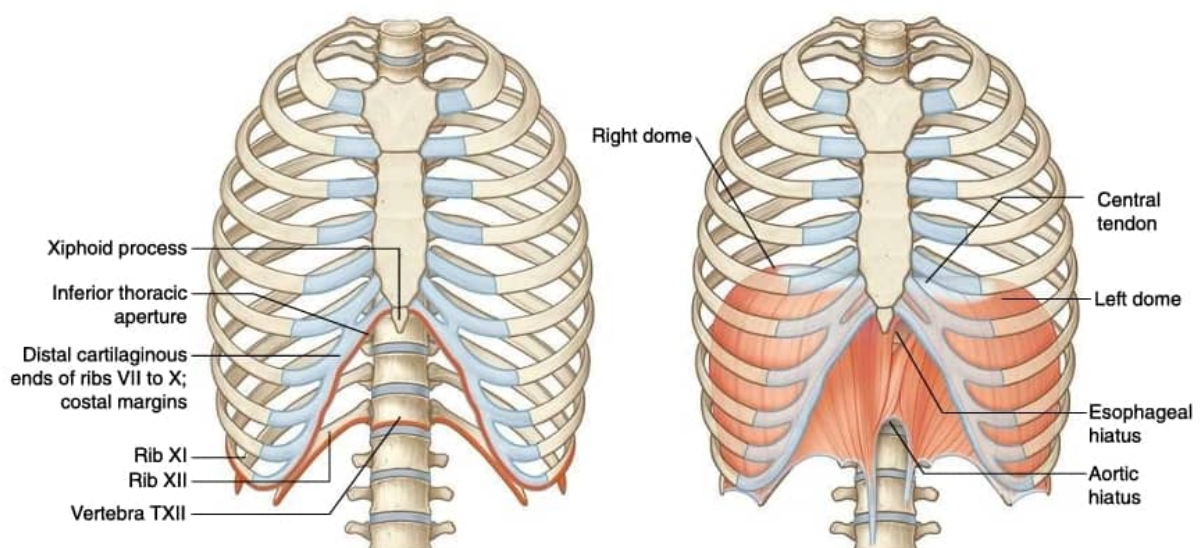
Suprapleural Membrane (Sibson's Fascia)

THORACIC OUTLET

- Its **plane** is sloping downwards and backwards.
- It is closed by **the diaphragm**

Boundaries

- **Anteriorly:** Xiphoid process
- **Posteriorly:** Body of T12 vertebra and 11th & 12th pairs of ribs
- **Anterolaterally:** Costal cartilages of 7th, 8th, 9th & 10th ribs





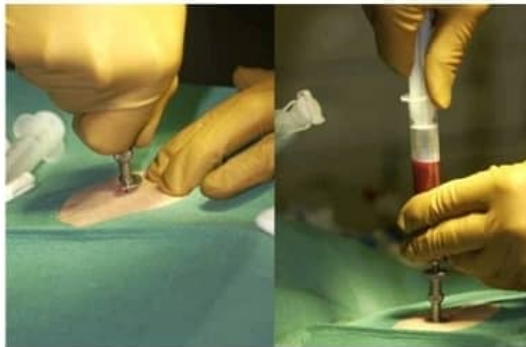
Clinical Note

1. Sternal puncture: A specimen of bone marrow (biopsy) can be obtained from the sternum as:

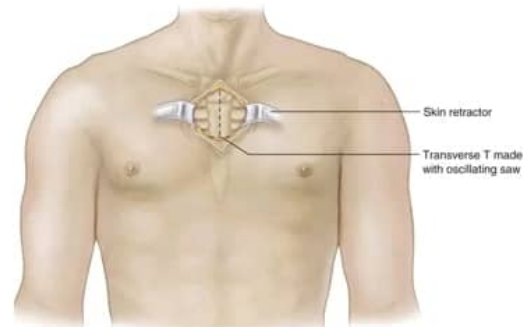
- a) It **contains red marrow** throughout life.
- b) It is **subcutaneous** and has a thin layer of cortical bone.
- The procedure is performed under local anesthesia using a widebore needle.

2. Sternal splitting:

- **The whole sternum** may be split **longitudinally** in modern **cardiac surgery** to gain access to the heart.
- **The manubrium** can be split in the **middle** to reach the superior mediastinum in operations on **retrosternal thyroid gland**.



Sternal puncture

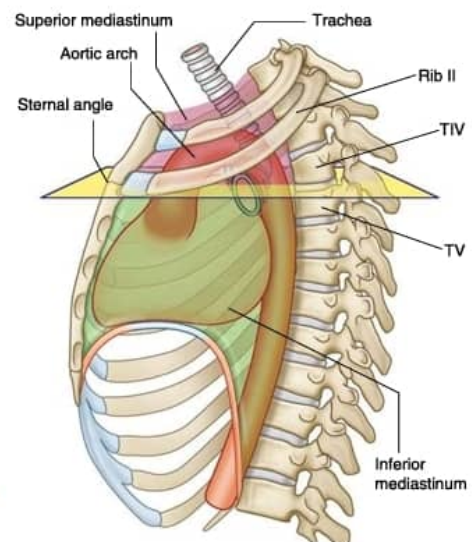
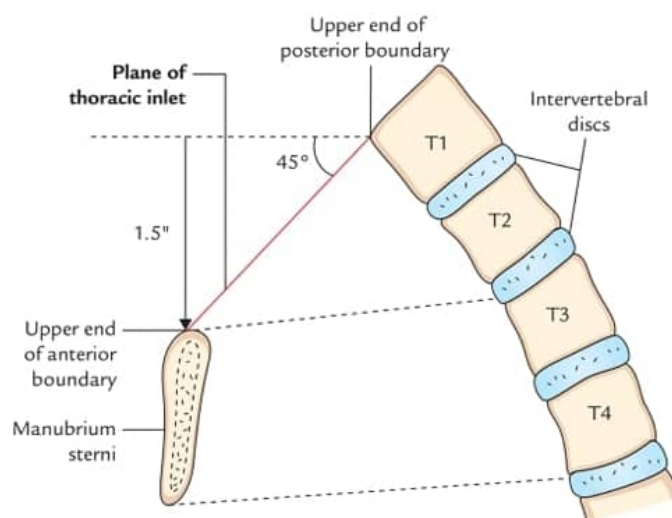


sternal splitting



Transverse Planes & Levels of the Thorax

- 1. The plane of the inlet of the thorax:** oblique, extending from the upper border of the manubrium (in front) to the upper border of the body of the 1st thoracic vertebra (behind).
- 2. The upper border of the manubrium (suprasternal or jugular notch):** lies opposite the lower border of the body of T2 vertebra or disc between T2/T3.
- 3. The plane of the sternal angle:** extends between manubrio-sternal junction and the lower border of T4 or disc between T4/T5.
- 4. Xiphisternal junction and xiphoid process:** lie opposite body of T9.



PARANASAL SINUSES

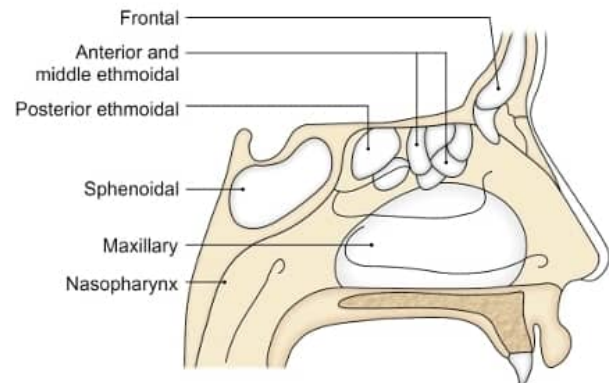
- **Definition:** They are air-filled spaces in the skull bones surrounding the nose.

- **Formation:** They are formed by **absorption of the spongy bone** present between the inner and outer tables of the skull bones.

- **Lining:** They are lined with **mucous membrane** which is continuous with that of the nose
 - They open in the lateral wall of nasal cavity by special foramina.

- **Functions:**

1. **Decrease** the weight of the skull due to filling of the sinuses by air.
2. **Increase** the resonance of voice.
3. **Increase** the surface area of nasal mucosa, so they warm and moist the inspired air.

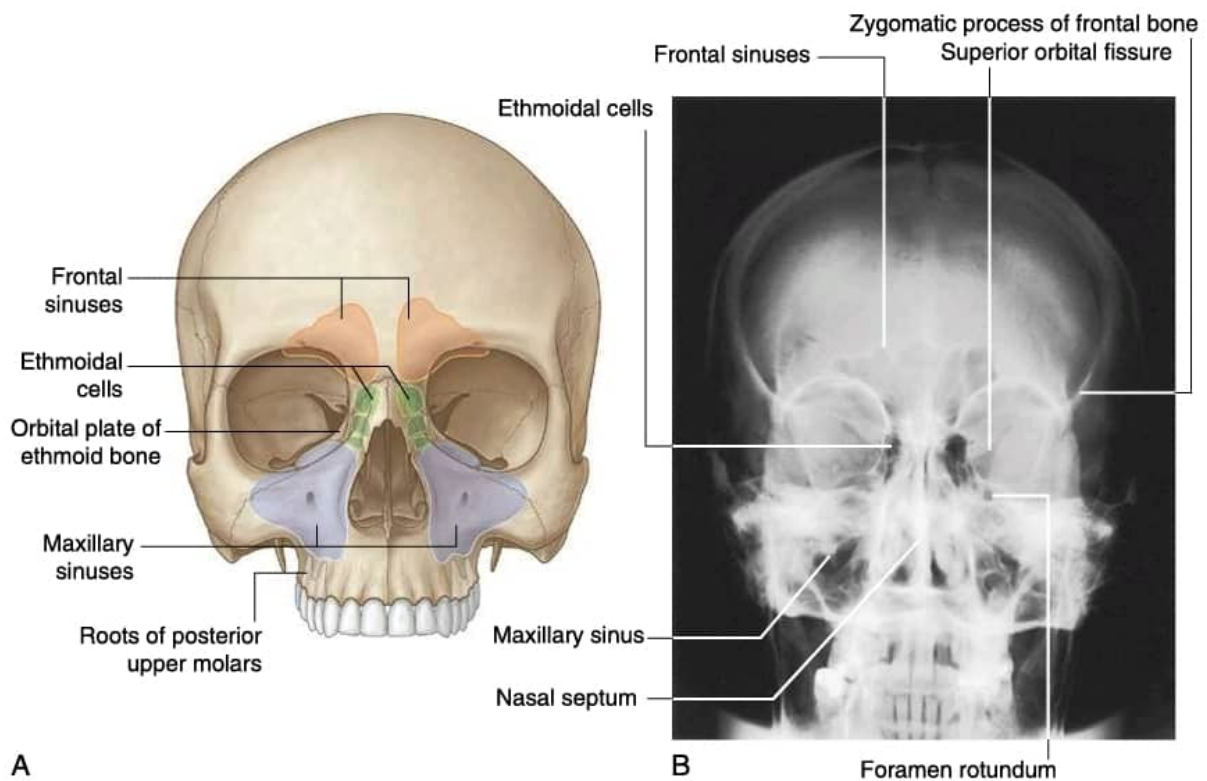


Wall of nasal cavity with location of paranasal sinuses

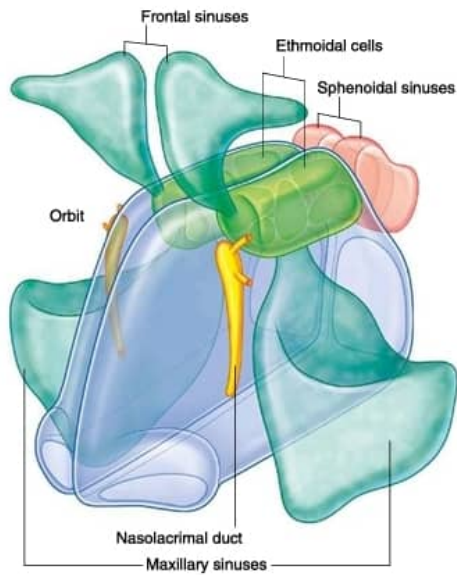


Important Note

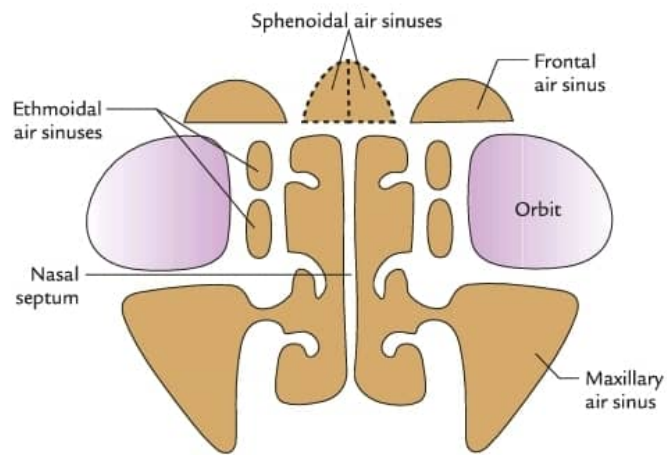
- The paranasal sinuses are absent at birth (**except** the maxillary sinus).
- They show marked growth at the time of **eruption of permanent teeth**.



Paranasal sinuses. A. Anterior view. B. Posteroanterior skull radiograph.



Paranasal sinuses and nasolacrimal duct.



Schematic diagram to show the location of various paranasal sinuses in relation to the orbit.

THE TYPES OF PARANASAL SINUSES

1. FRONTAL SINUSES

- Site:

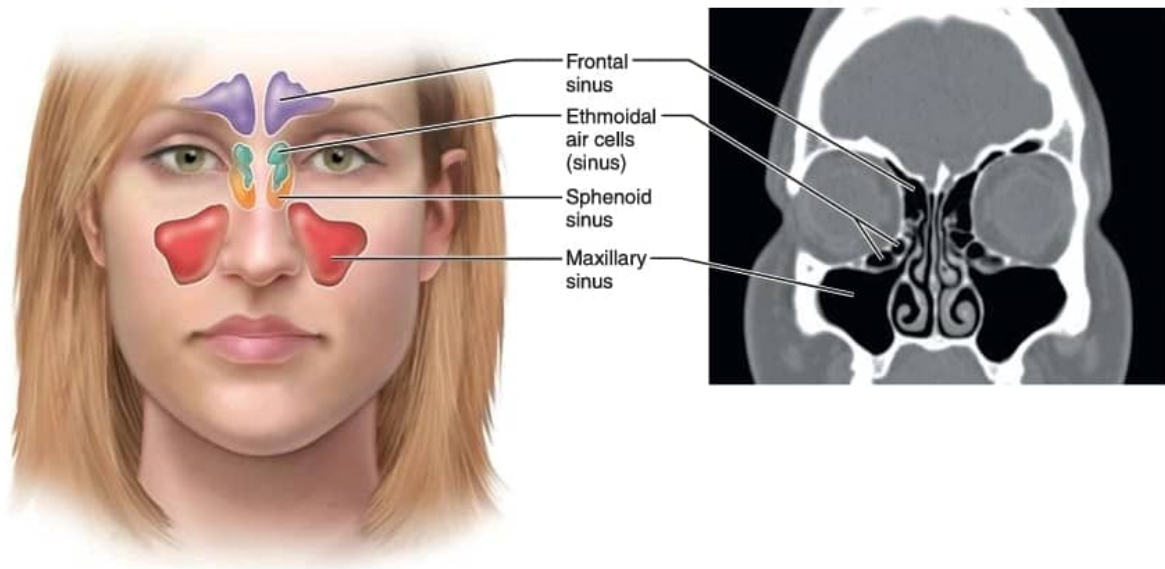
- They are two asymmetrical sinuses present in the **squamous part of the frontal bone** above & behind the **superciliary arches**.
- They are separated by a bony septum which is usually deviated to one side.

- **Opening:** Each sinus opens into the **frontonasal duct (ethmoidal infundibulum)** which opens in the **anterior end of the hiatus semilunaris** in the middle meatus of nose.

- **Nerve supply:** Supraorbital and anterior ethmoidal nerves.

- **Arterial supply:** Supraorbital artery.

- **Lymph drainage:** Into the submandibular lymph nodes.



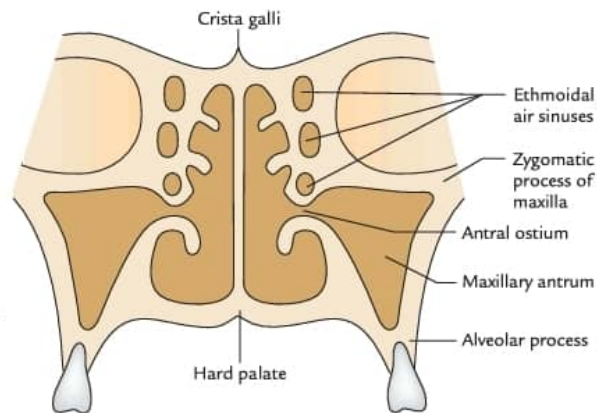
Paranasal sinuses. Illustration on left, CT scan on right.

2. MAXILLARY SINUSES

- It **the largest** air sinus and **the only** one present at birth.
- It is pyramidal in shape and occupies the whole of the body of maxilla.

Relations

- **Apex:** It's directed laterally **towards the zygomatic bone**.
- **Base:** It's directed medially **towards the nasal cavity**.
- **Roof:**
 - **The floor of the orbit**, containing infraorbital groove & canal and infraorbital nerve & vessels
 - It separates the sinus from **the contents of the orbit**.
- **Floor:**
 - It's formed by **alveolar process of maxilla**.
 - It lies 1 cm below the floor of the nose.
 - It's lower part lies opposite **the 2nd premolar and 1st molar teeth**.
- **Anterior wall:** It's **subcutaneous**, containing anterior superior alveolar nerve and vessels.
- **Posterior wall:**
 - It separates the sinus from **infratemporal and pterygopalatine fossae**.
 - It is pierced by posterior superior alveolar nerve and vessels.



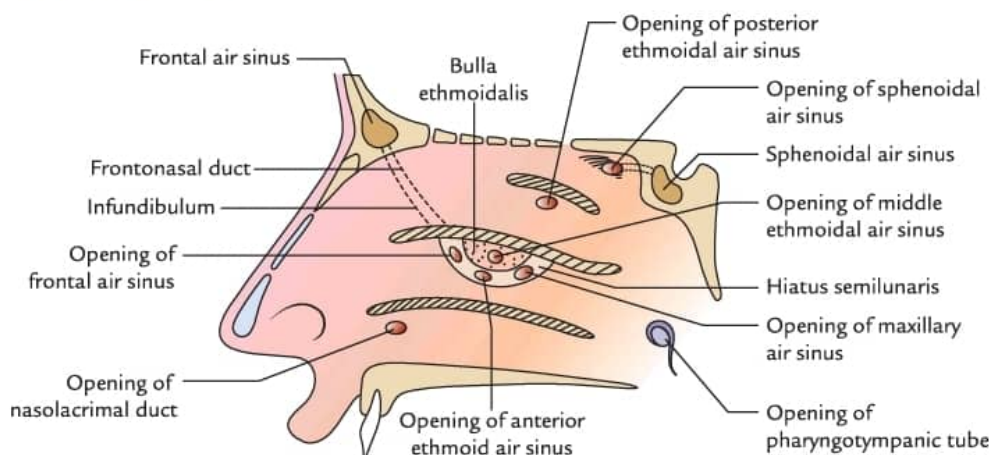
Relations of maxillary sinus.

- **The extraction of the 2nd premolar & 1st molar teeth** may damage and perforate the floor leading to **oroantral fistula**.

Openings of maxillary sinus

- It's large rounded opening, 2 cm in diameter
- It's present **in the upper part** of the base of the sinus.
- **Site of opening:** It opens in the middle meatus of the nose at **the posterior end of hiatus semilunaris** below the bulla ethmoidalis.
- **Accessory small openings** are usually present behind the main opening.

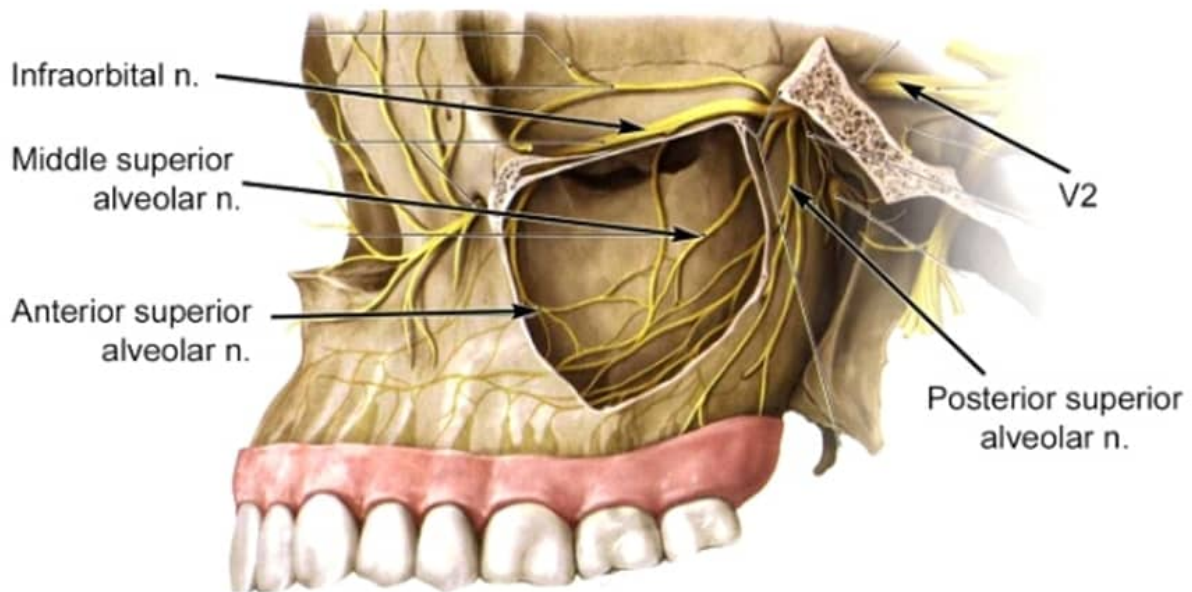
- **The higher position of the opening of the sinus** makes the drainage of any collection of pus or blood **very hard**.



Lateral wall of the nose with conchae removed showing openings of various sinuses and nasolacrimal duct.

Nerves & Vessels related to the maxillary sinus

1. **Anterior superior alveolar:** A branch from **infraorbital** nerve and vessels.
 - They descend in **the anterior wall** supplying it.
2. **Middle superior alveolar:** A branch from **infraorbital** nerve and vessels.
 - They descend in **the lateral wall of the sinus** to supply it.
3. **Posterior superior alveolar:** A branch from **maxillary** nerve and vessels.
 - They descend and supply **the posterior wall of the sinus**.
4. **Infraorbital nerve and vessels:**
 - They pass in the infraorbital groove and canal through the roof of the sinus.
5. **Greater palatine nerve and vessels:**
 - They descend behind the sinus supplying **its posterior wall**.



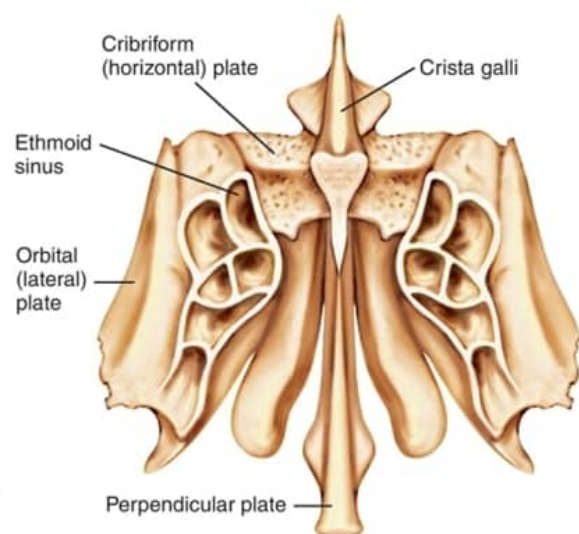
3. ETHMOIDAL SINUSES

- Site: They are intercommunicating spaces lying **inside the ethmoidal labyrinth** between the orbit and upper part of nasal cavity.

- Parts: They vary in number from 3-18 air cells arranged into 3 groups:

1. **Anterior ethmoidal sinus (or cells):** opens in the ethmoidal infundibulum or in the anterior end of the hiatus semilunaris in the middle meatus of nose.
2. **Middle ethmoidal sinus (or cells):** **produces** the bulla ethmoidalis on which it opens.
3. **Posterior ethmoidal sinus (or cells):** opens in the superior meatus.

- Nerve and arterial supply: Anterior and posterior ethmoidal nerves and arteries.





Ethmoid bone. A. Overall shape. B. Coronal section through skull.

4. SPHENOID SINUS

- Site:

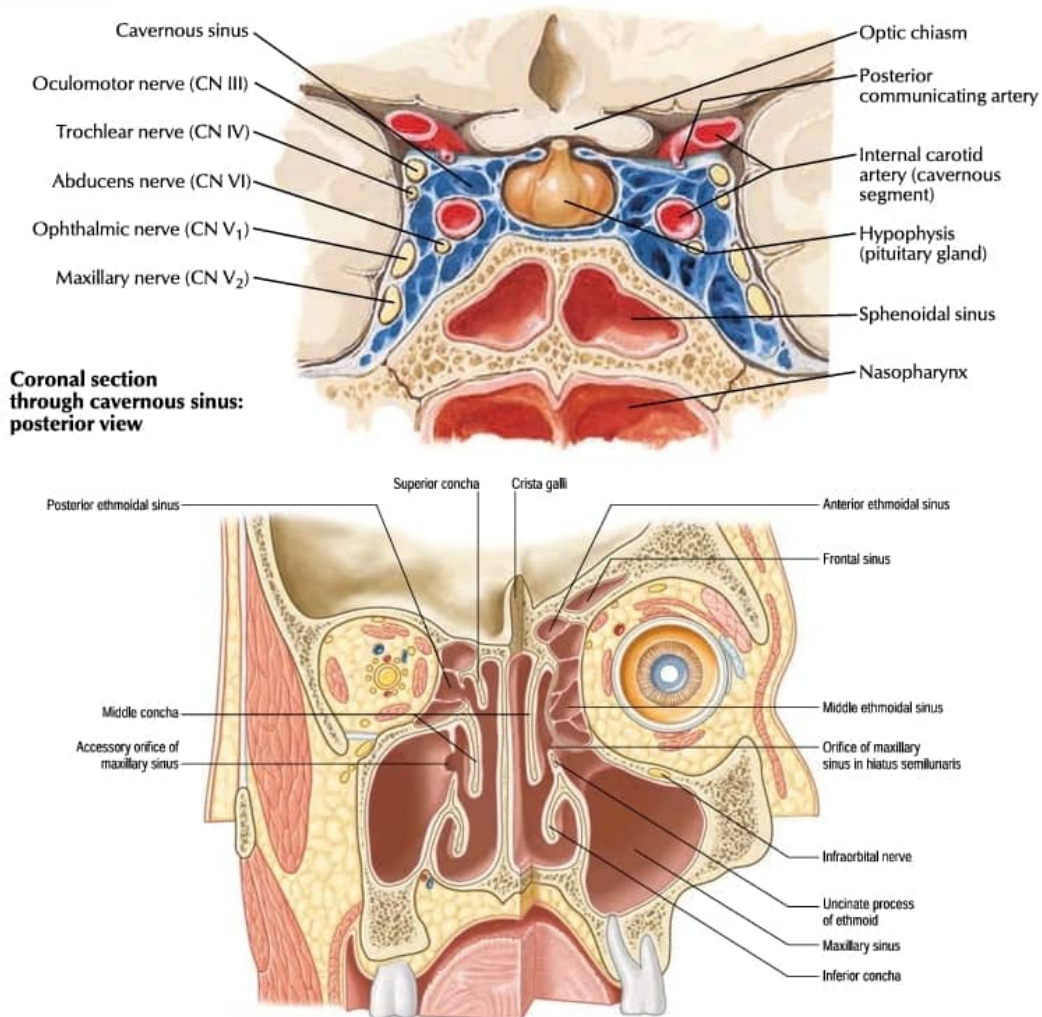
- It occupies the body of sphenoid & lies behind the upper part of nasal cavity.
- The two sphenoid sinuses are separated by a bony septum which lies in the midline or may be deviated to one side in most of the cases

- Relations:

- **Superiorly:** Pituitary gland and optic chiasma.
- **Posteriorly:** Pons and basilar artery.
- **Anteriorly:**
 - a. The opening of the sinus lies in the upper part of its anterior wall and it opens in the sphenothmoidal recess.
 - b. **The sphenopalatine foramen** (with structures passing through it) lies anterior to the lower part of the sinus.
- **Laterally:** On both sides there are **cavernous sinuses** with the internal carotid arteries and abducent nerves

- **Nerve supply:** Posterior ethmoidal branches of nasociliary nerve & orbital branches of the sphenopalatine ganglion.

- **Arterial supply:** Posterior ethmoidal artery of ophthalmic artery.

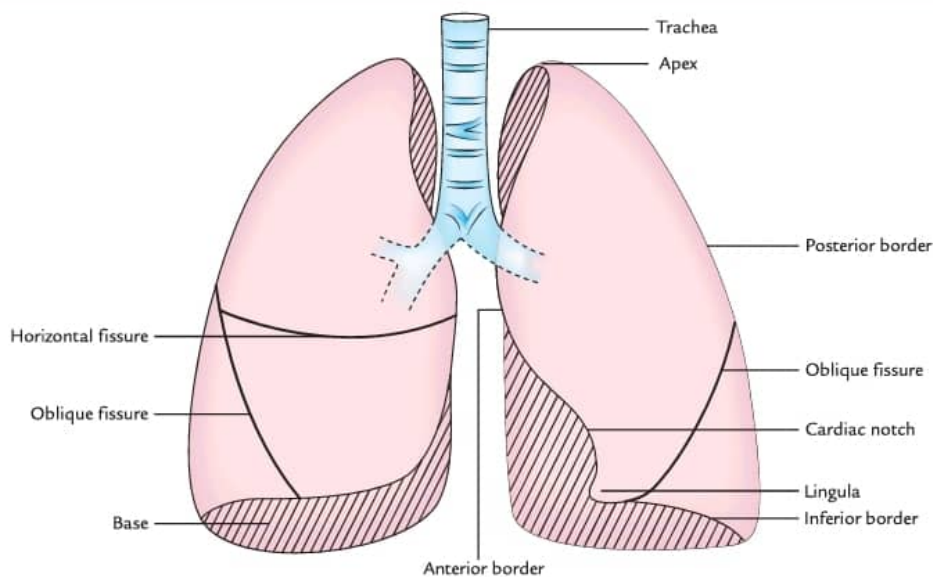


A coronal section through nasal cavity, viewed from posterior aspect. The plane of the section on the right side is more anterior. The normal orifice of the maxillary sinus is shown on the right side and an accessory orifice on the left side.



LUNGS

- The lungs are the organs of gaseous exchange between the blood and the atmospheric air.
- **Before birth**, the lung is **firm** and **sinks** when put in water.
- Each lung resembles **half a cone**.
- **Parts:** It has an apex, a base, 2 surfaces (costal & medial) and 3 borders (anterior, posterior & inferior).



A. Apex of the Lung

- **Site:** Rounded, upper end which projects upwards, **through the thoracic inlet**, into the root of the neck.
 - It reaches the neck of the 1st rib (**behind**) but lies about one inch above the medial third of the clavicle (**in front**).
- **Relations:**
 - **Above and in front:** the apex is covered by the cervical pleura and suprapleural membrane. It is grooved by the subclavian artery arching upwards and laterally.
 - **Behind:** the apex is separated from the neck of the 1st rib by the three structures which lie in front of the neck

B. Base of the Lung

- **Site:** It is the concave, lower surface which rests on **the cupola of the diaphragm**.

- **Relations:** The base of the lung is separated by the diaphragm from:

- Right lobe of the liver (**right lung**)
- Left lobe of the liver, fundus of the stomach and spleen (**left lung**).

C. Surfaces of the Lung

1. **Costal surface:** wide, smooth & convex.

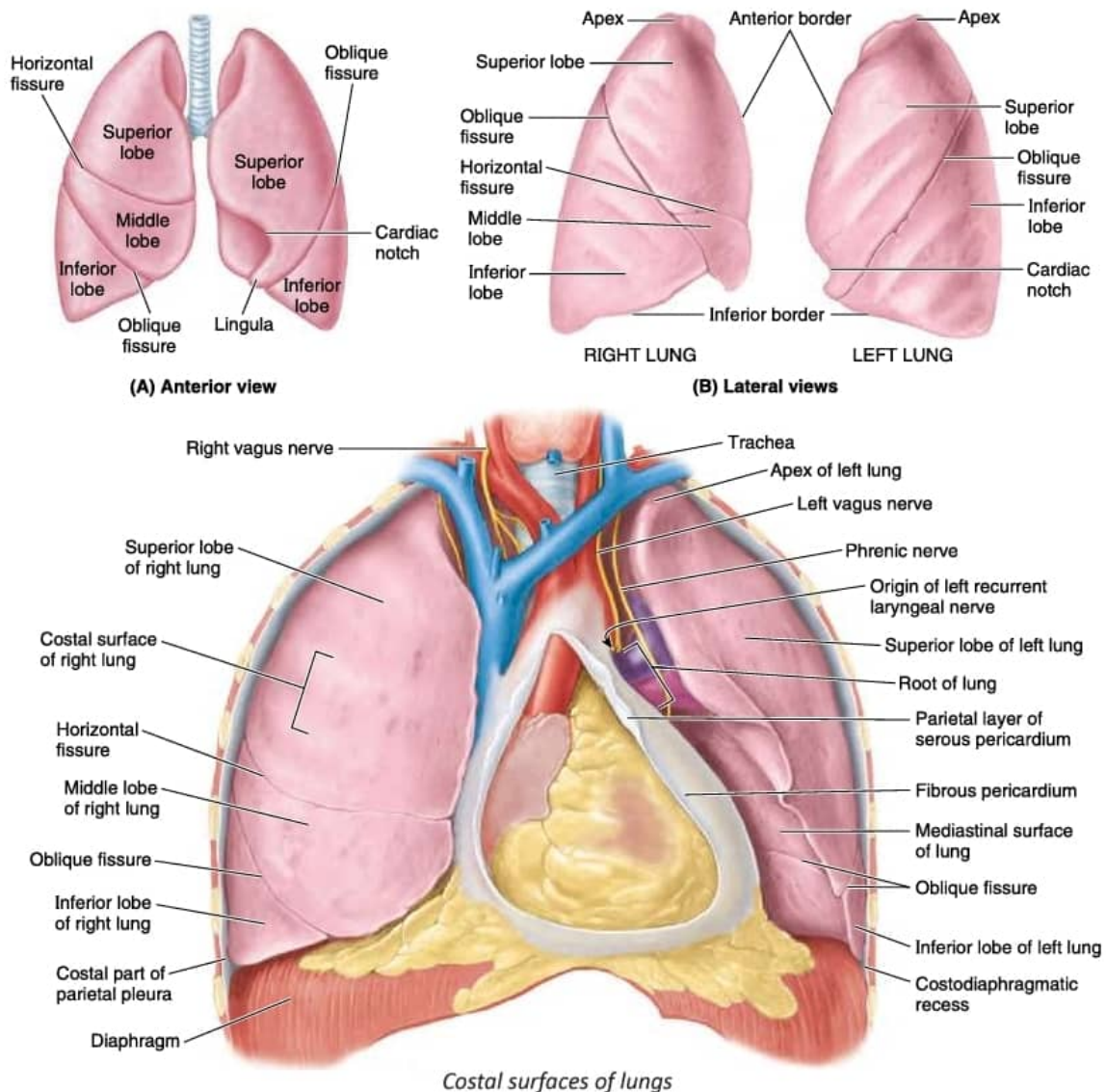
- It is **related to** the ribs, costal cartilages & intercostal spaces.

2. **Medial surface:** marked by the presence of the hilum of the lung. It is divided into 2 parts:

a. **Vertebral (posterior part):** has same relations in both lungs to:

- Bodies of thoracic vertebrae and intervertebral discs.
- Intercostal nerves and posterior intercostal vessels.
- Sympathetic chain and splanchnic nerves.

b. **Mediastinal (anterior part):** presents the hilum of the lung and several grooves & impressions which differ between the right and left lungs.



D. Borders of the Lung

1. Anterior border: thin and sharp.

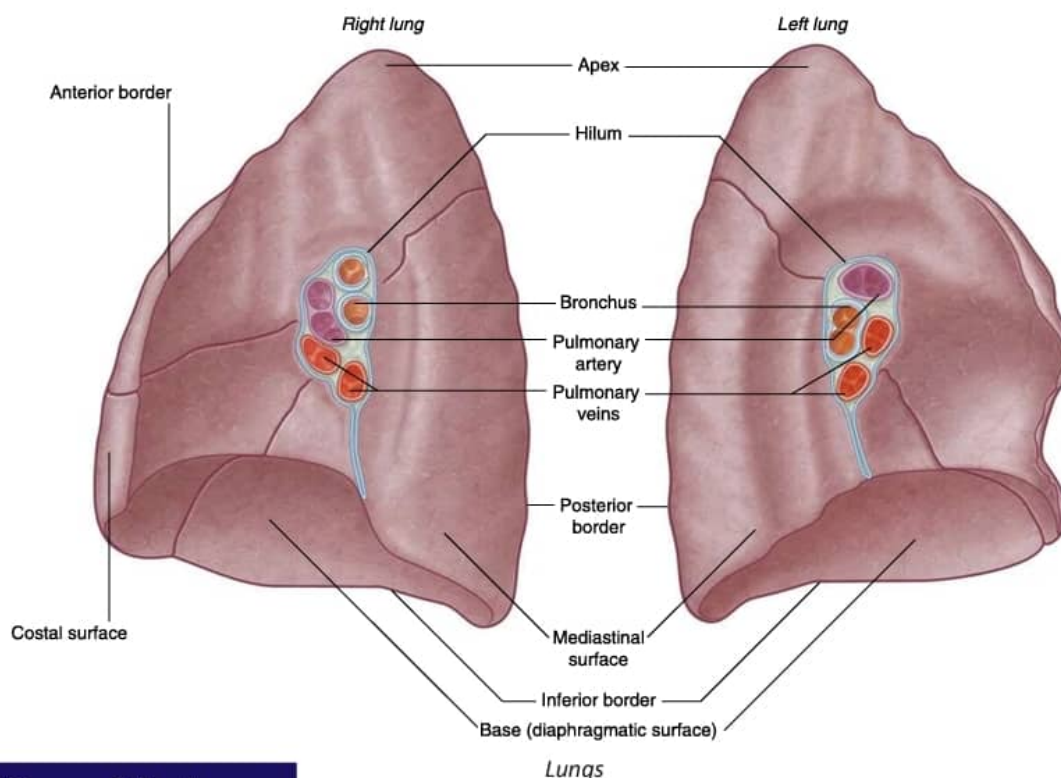
- It separates the costal surface from the mediastinal part of the medial surface.
- It has a **notch** in its upper part caused by the 1st rib.
- The anterior borders of the right and left lungs are different:
 - a. In the right lung**, the anterior border is straight and corresponds to the anterior border of the pleura.
 - b. In the left lung**, the lower part of the anterior border presents the cardiac notch followed by a tongue-like process, **the lingula**.

2. Inferior border: encircling the diaphragmatic surface.

- It forms the boundary of the base of the lung.

3. Posterior border: thick and rounded

- It separates the costal surface from the vertebral part of the medial surface.



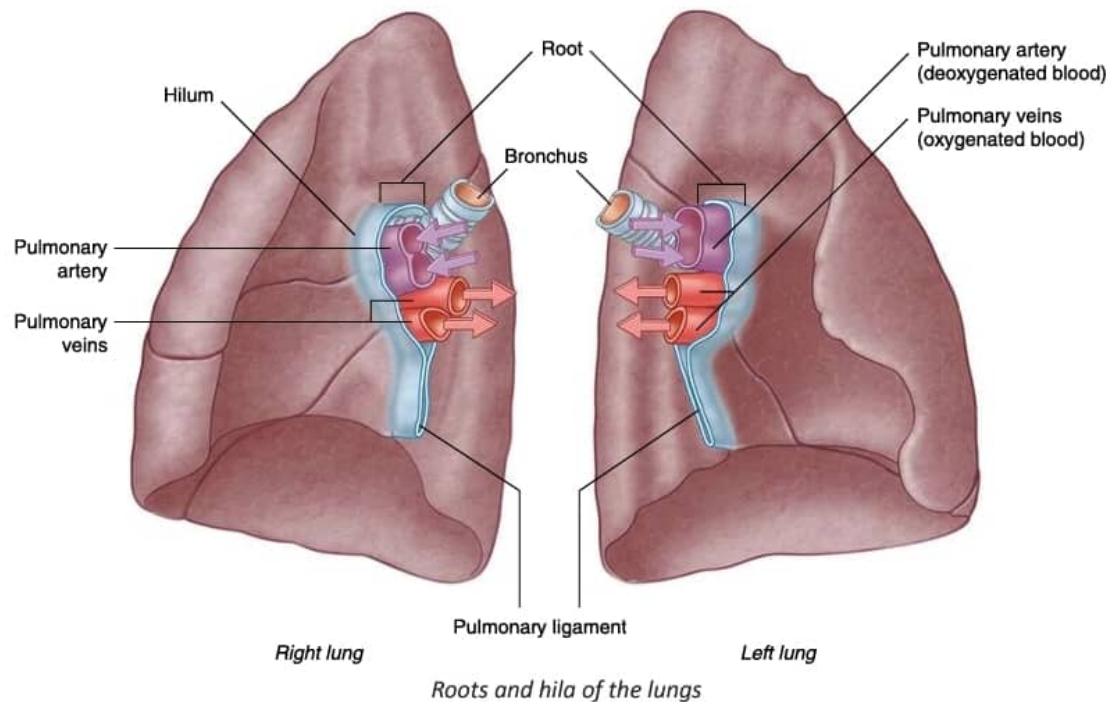
Hilum of the Lung

A- In the right lung:

1. Inferior pulmonary vein (**most inferiorly**).
2. Superior pulmonary vein (**most anteriorly**).
3. Principal (or hyparterial) bronchus (**most posteriorly**).
4. Pulmonary artery (**between principal bronchus and superior pulmonary vein**).
5. Superior lobar (eparterial) bronchus (**above principal bronchus**).
6. Branch from pulmonary artery (**in front of eparterial bronchus**).

A- In the left lung:

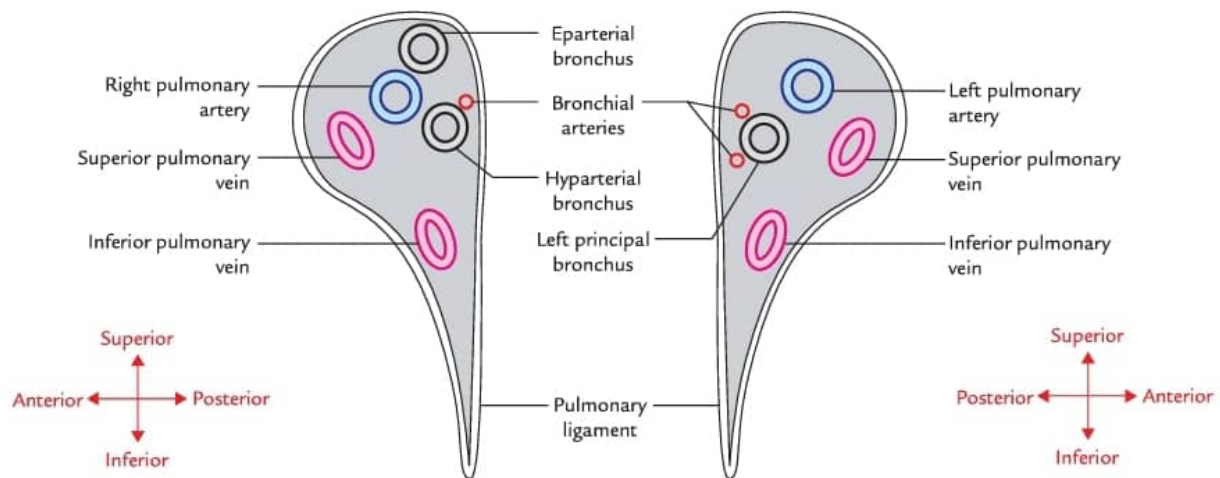
1. Inferior pulmonary vein (**most inferiorly**).
2. Superior pulmonary vein (**most anteriorly**).
3. Principal bronchus (**most posteriorly**).
4. Pulmonary artery (**above and in front of principal bronchus**).



Root of the Lung

Contents of the Root of the Lung

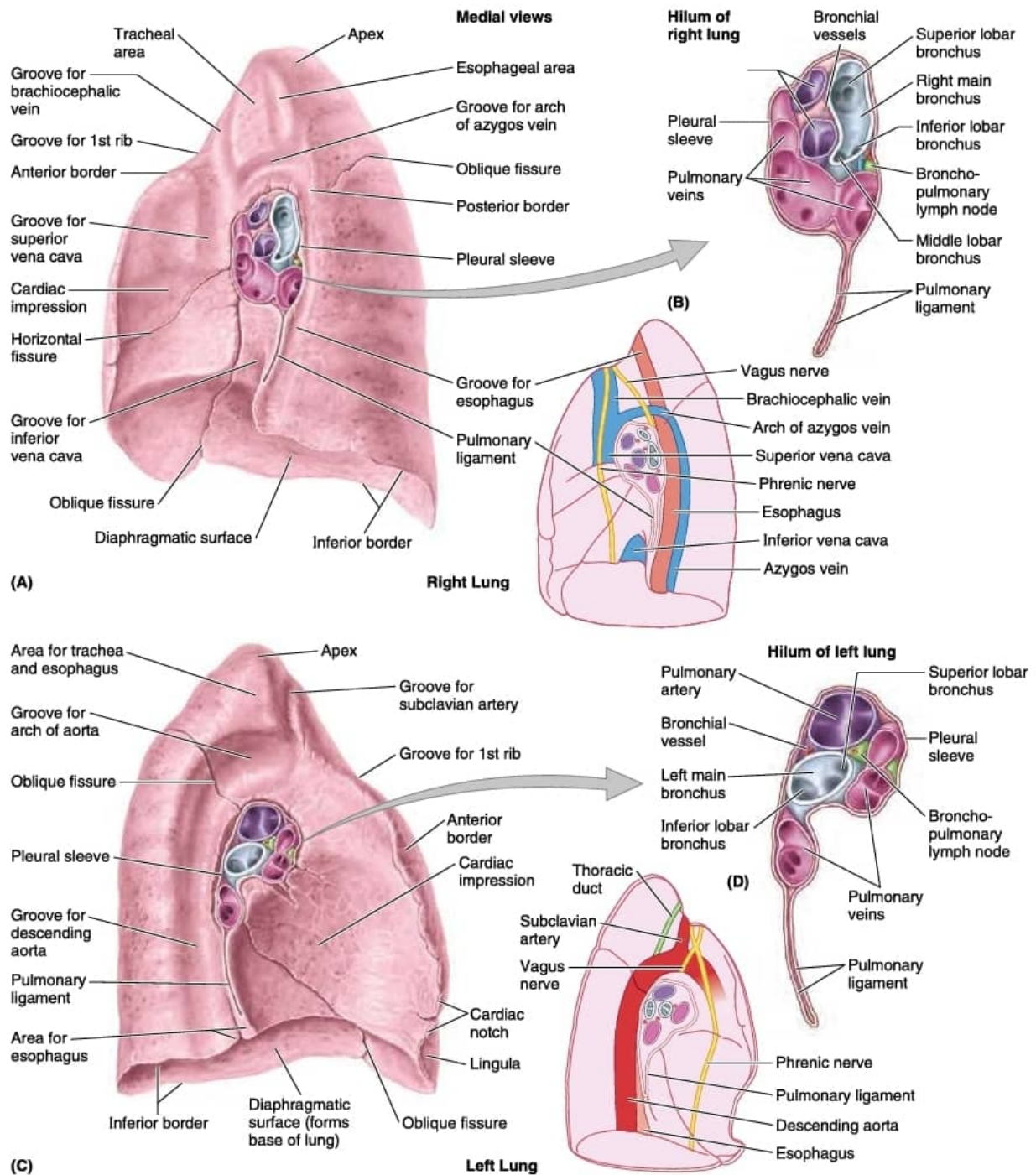
1. **Inferior pulmonary vein** (most inferiorly).
2. **Superior pulmonary vein** (most anteriorly).
3. **Principal bronchus** (most posteriorly).
4. **Pulmonary artery:**
 - a. **In the right root:** (between principal bronchus and superior pulmonary vein)
 - b. **In the left root:** (above and in front of the principal bronchus).
5. **Bronchial vessels:** along the posterior wall of the bronchi;
 - a. **The right lung** receives one bronchial artery
 - b. **The left lung** has two.
6. **Bronchopulmonary lymph nodes and lymph vessels:** within the root.



Arrangement of structures in the roots of right and left lungs.

Relations of the root of the lung

- **In front:** phrenic nerve and anterior pulmonary plexus.
- **Behind:** vagus nerve (breaks up to form posterior pulmonary plexus), oesophagus (**on the right**) and descending thoracic aorta (**on the left**).
- **Above:** arch of azygos vein (**on the right**) and arch of aorta (**on the left**).
- **Below:** pulmonary ligament.



HISTOLOGY SECTION



SECTION II





RESPIRATORY SYSTEM

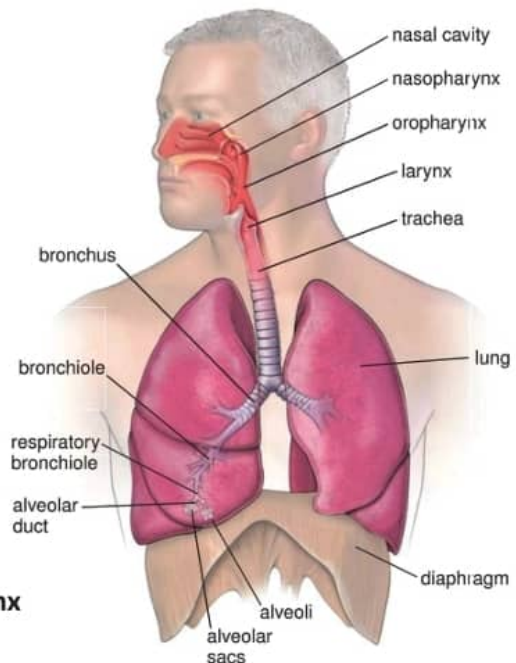
- The respiratory system consists of two lungs and numerous tubes that link to external environment.
- It is responsible for the intake of oxygen in the inspired air and elimination of carbon dioxide produced by the metabolism of cells throughout the body
- The respiratory system is divided into:
 - I- Conducting portion.
 - II- Respiratory portion.

I) CONDUCTING PORTION

Main Functions

- It serves two functions:
 - 1) Conduction of air:** It is provided with:
 - **Cartilage and collagen fibers.** Why?
 - a. To prevent collapse
 - b. To ensure an uninterrupted supply of air.
 - **Elastic fibers and smooth muscles** for flexibility and extensibility.
 - 2) Conditioning of inspired air:**
 - Air is cleansed, moistened and warmed by the action of:
 - a. **Respiratory epithelium** (pseudostratified columnar ciliated with goblet cells).
 - b. **The vascular network, mucous and serous glands** in the C.T. under the epithelium.
 - It includes:

1) Nasal cavity	2) Nasopharynx
3) Larynx	4) Trachea
5) Bronchi.	6) Bronchioles
7) Terminal bronchioles	



RESPIRATORY EPITHELIUM

- The epithelium that lines most of the conducting portion is **pseudostratified columnar ciliated with goblet cells**.
- Epithelial cell types: **Five cell types** are recognized:

1- Ciliated columnar cells

- **The most abundant cells.**
- They are **columnar ciliated cells** (300 motile cilia/cell).
- **The apical cytoplasm** contains basal bodies and numerous mitochondria which supply energy for ciliary beating.
- **Function:** They **push the mucus** with the trapped bacteria and dust in one direction outwards.

2- Mucous goblet cells

- **The next most numerous.**
- **Their apical portions** are filled with mucin granules.
- **Function:** They **secrete mucus** that covers the epithelium and traps bacteria & dust particles.

3- Brush cells

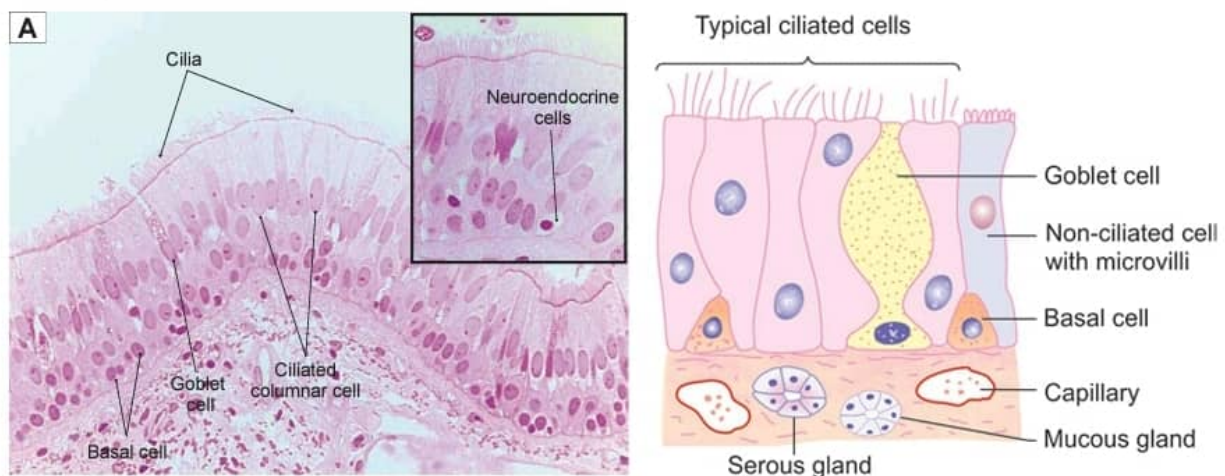
- They are **columnar cells** with abundant apical microvilli.
- They have **basal afferent nerve endings**.
- **Function:** They act as **sensory receptors**.

4- Basal cells

- They are **small rounded cells** on the basal lamina but **do not** reach the lumen.
- **Function:** They are **stem cells** that replace other cell types.

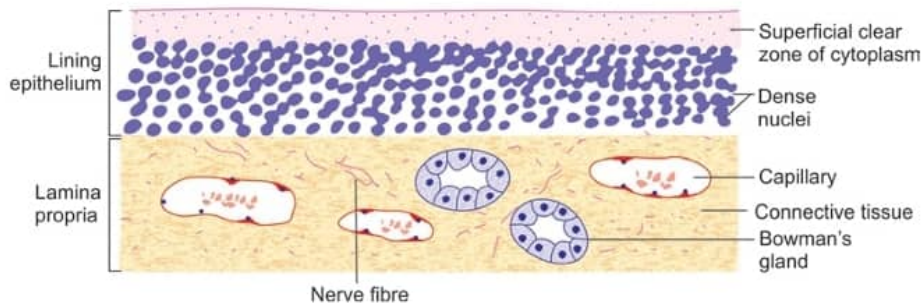
5- Small granule cells

- They are **neuroendocrine cells belonging to APUD**.
- They have numerous **basal dense core granules**.
- **Function:** They secrete **hormones like serotonin & catecholamines** that:
 - a. Control the diameter of airways & blood vessels
 - b. Regulate mucous & serous secretions.



B) Lamina propria

- It is a **dense fibroelastic C.T.** that contains blood capillaries and olfactory nerve bundles.
- It contains (**Bowman's glands**) that secrete **serous fluid** that passes through ducts to the surface.
 - This fluid creates an environment around the olfactory cilia that acts as a **solvent for the odorous substances.**



Clinical Note

- **Olfactory neurons** are considered as **the only neurons** that can highly regenerate due to high capacity of stem cells.

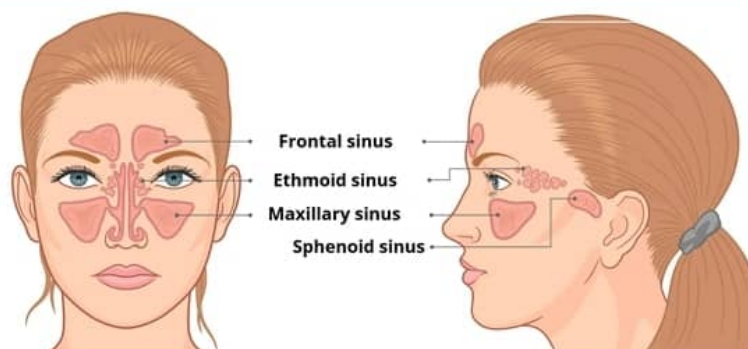
DIFFERENCES BETWEEN RESPIRATORY & OLFACTORY MUCOSA

	OLFACTORY MUCOSA	RESPIRATORY MUCOSA
Epithelium	Thicker	Thinner
Goblet cells	Absent	Present
Cilia	Few long non motile cilia	Numerous true cilia
Basement membrane	Thin	Thick
Bowman's glands	Present	Absent
Pigment	Present in sustentacular cells	Absent



Paranasal Sinuses

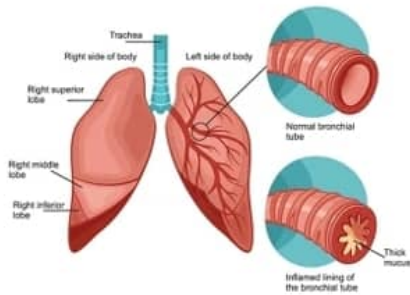
- **Definition:** They are paired closed cavities in the frontal, maxillary, ethmoid & sphenoid bones of skull.
- **Lining:** They are lined with thinner respiratory epithelium with few goblet cells.
- **Sinusitis:** is an inflammatory process of the sinuses which may be acute or chronic.





Clinical Note

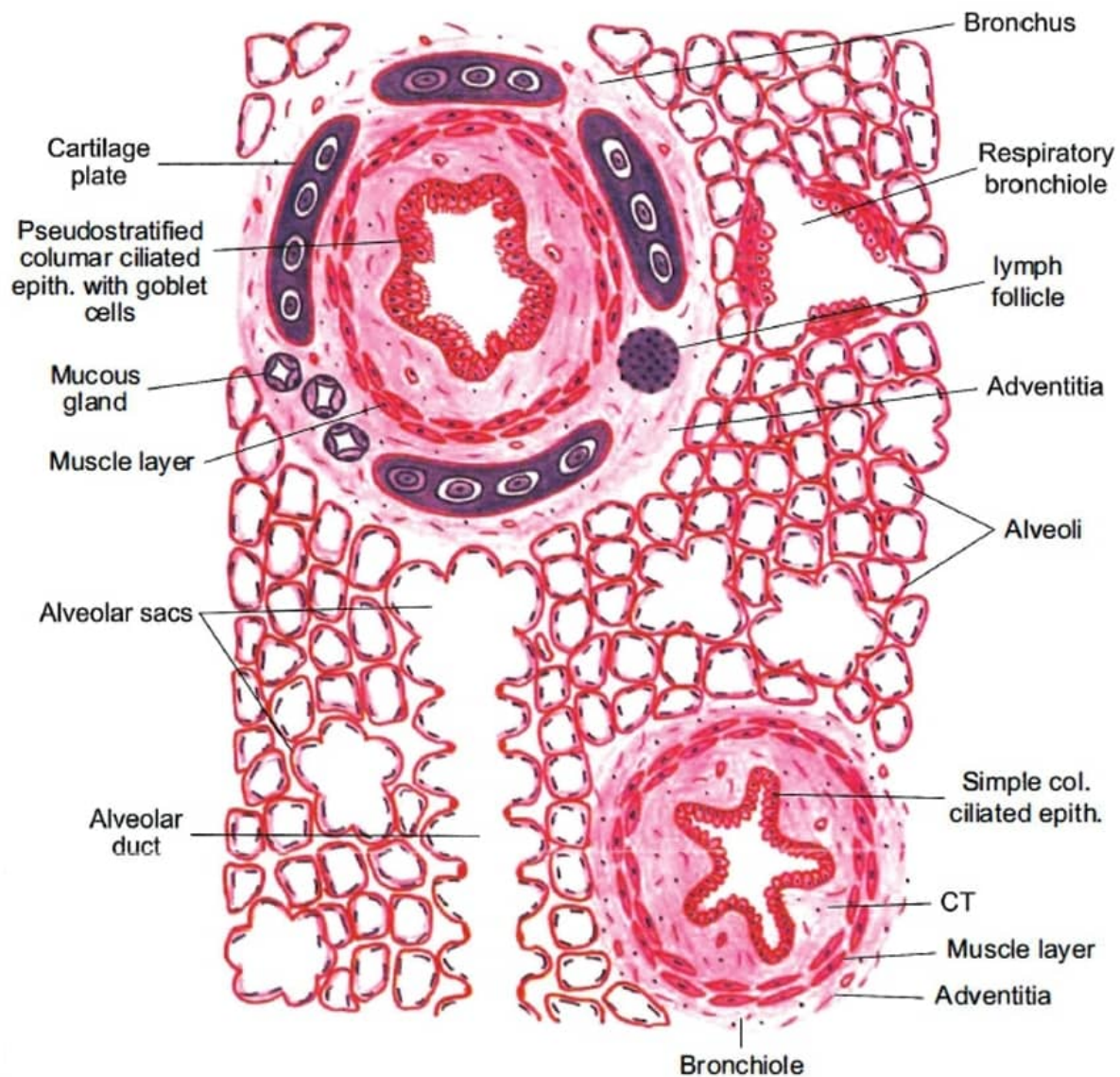
- **Bronchitis**: is inflammation of the mucous membrane of the bronchi which appears congested with swollen mucous glands.
- **Bronchial asthma**: is an allergic condition, in which spasm of bronchiolar smooth muscle causes constriction of bronchioles and difficulty during expiration.



Bronchitis



Bronchial asthma



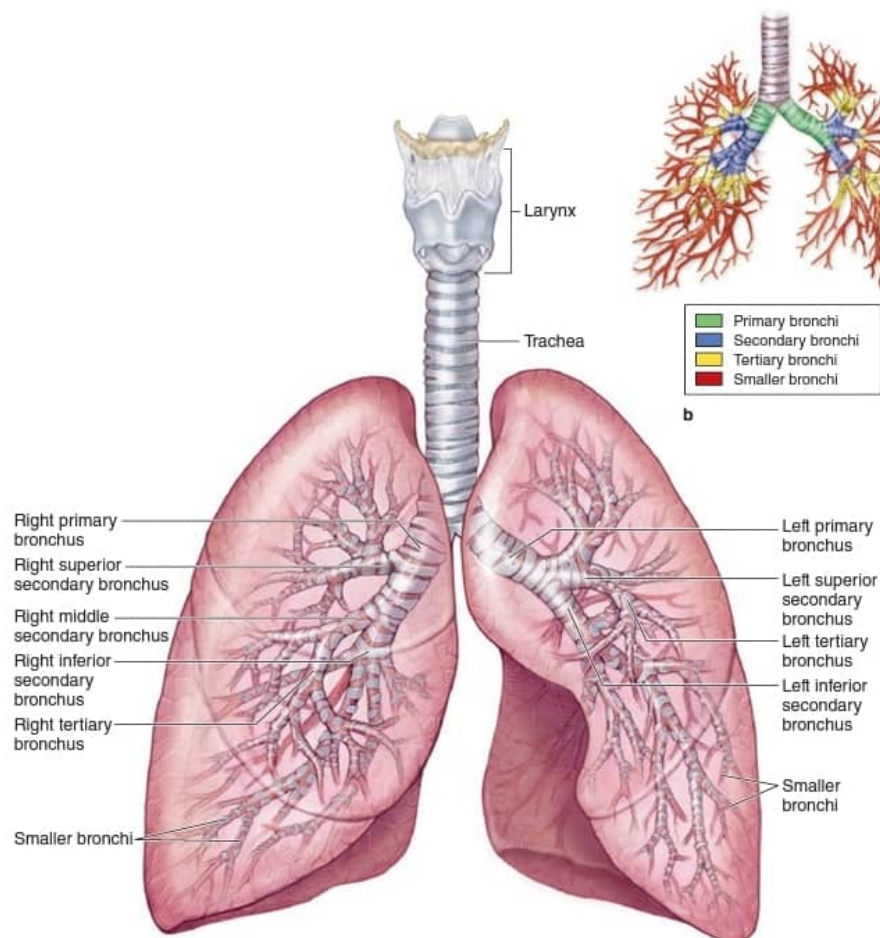
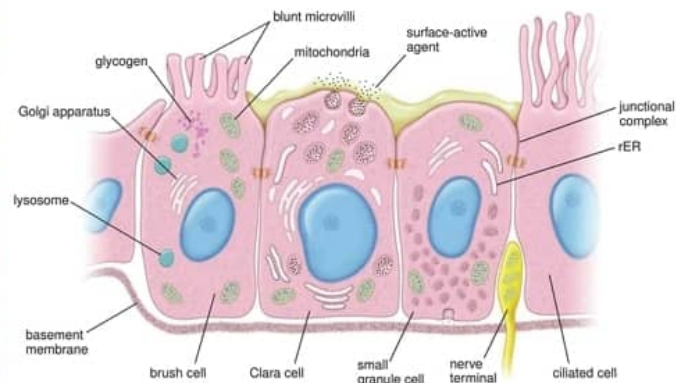
CLARA CELLS

- **L.M.:** Dome-shaped, non-ciliated cells.

- **E.M.:** basal rER, apical sER, mitochondria, Golgi apparatus and glycoprotein granules.

- **Functions of Clara cells:**

1. **Protection of bronchiolar epithelium** via glycoproteins in their granules (**rER&GA**).
2. **Degradation of toxins** in the inhaled air (**sER**).
3. **Secretion of a surfactant like material** which reduces surface tension of bronchioles and maintains their patency (**sER&GA**).
4. **Protection against emphysema** by inhibiting proteases & elastases secreted by macrophages.
5. **Stem cells** for other types.



PHYSIOLOGY SECTION



SECTION III



RESPIRATORY PRESSURES

- Air flow into and out of the lungs (**breathing or ventilation**) occurs because of the presence of **pressure gradients** between the alveoli and the atmosphere.
- **Bulk flow can be described by the equation: $F = \Delta P / R$**

1. ATMOSPHERIC PRESSURE (P_{atm})

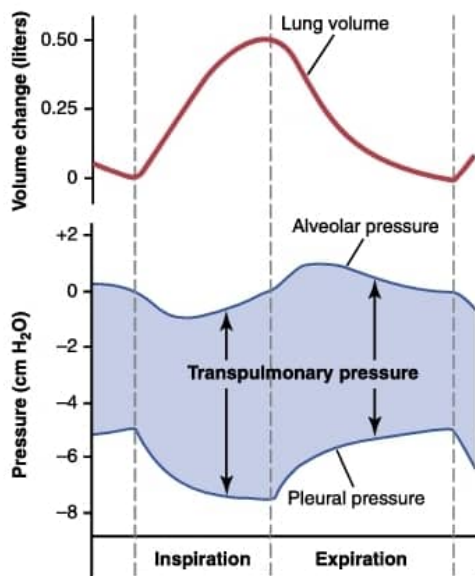
- **At sea level**, atmospheric pressure is normally **760 mmHg**.
- For simplicity, we assume that atmospheric pressure is constant.
- All other pressures in the respiratory system are expressed relative to atmospheric pressure, i.e., "**zero reference pressure**".

2. INTRA-ALVEOLAR PRESSURE (P_{alv} OR P_A)

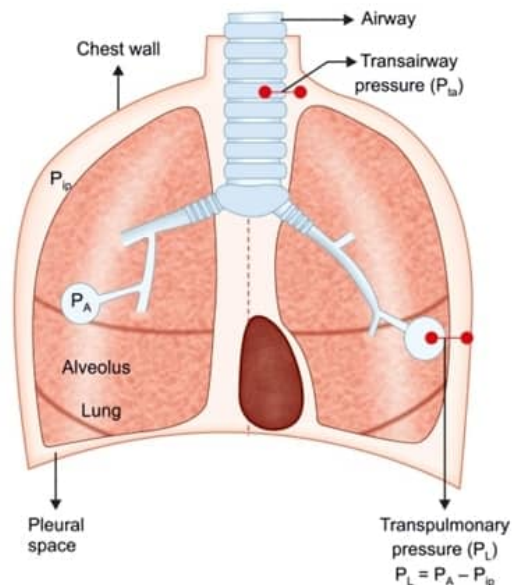
(INTRA-PULMONARY PRESSURE)

- **It is the pressure inside alveoli.**

1. **Before inspiration:** when the glottis is opened and **no air** is flowing in or out of the lungs, the pressure in all parts of the respiratory tract down to the alveoli is **equal to atmospheric pressure**, and is considered **zero pressure**.
2. **During inspiration** with **expansion** of the chest, alveolar pressure **falls below** the atmospheric pressure (**to about $-1 \text{ cm H}_2\text{O}$**).
 - Thus, a pressure gradient between the atmosphere and the alveoli is created, and air **moves into** the lungs.
3. **At the end of inspiration** the intra-alveolar pressure **returns to zero**.
4. **During expiration** opposite changes occur in the lungs.
 - The chest wall **recoils** to its normal position.
 - **However**, their content of air has increased by the volume of air entered during inspiration.
 - This **increases** the alveolar pressure to **$+1 \text{ cm H}_2\text{O}$** .
 - A pressure gradient in the opposite direction results, and air **moves out** of the lungs.
5. **At the end of expiration** the intra-alveolar pressure **returns to zero**.



Changes in lung volume, alveolar pressure, pleural pressure, and transpulmonary pressure during normal breathing.

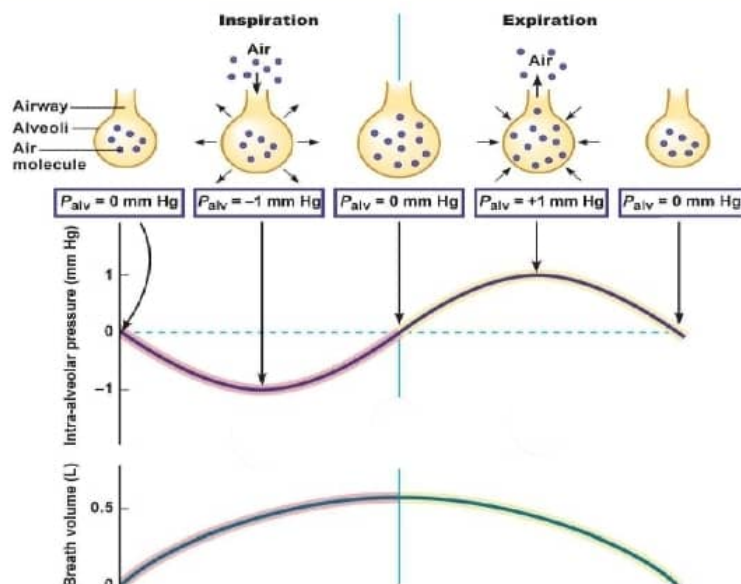


Alveolar (P_A), transairway (P_{ta}) and transpulmonary (P_L) pressures.



Important notes about intra-alveolar pressure

- P_{alv} is **zero** at: • Beginning of inspiration. • End of inspiration. • End of expiration.
- P_{alv} is **negative** during inspiration and becomes **positive** during expiration.



Changes in intra-alveolar pressure during normal quiet breathing: 1. before inspiration, 2. during inspiration, 3. at end of inspiration, 4. during expiration and 5. at the end of expiration.

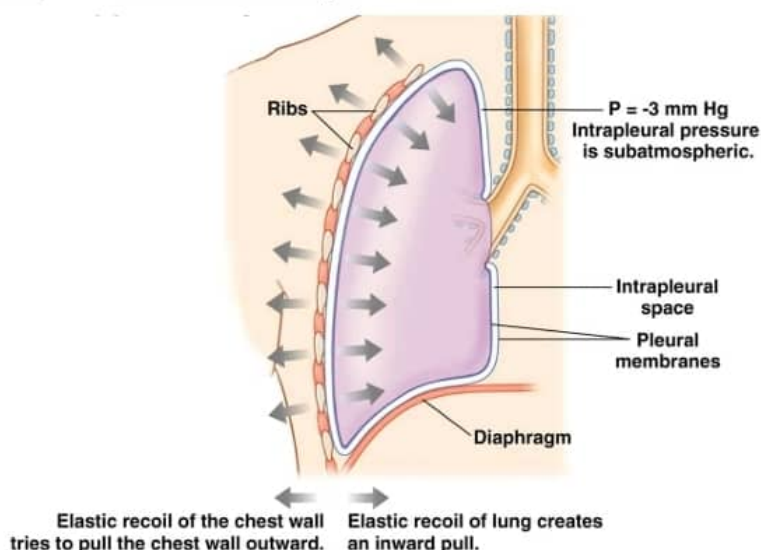
3. INTRA-PLEURAL PRESSURE (P_{PI} OR P_{ip})

- It is the pressure in the space between visceral and parietal pleura.
- The intrapleural pressure is always **negative** under normal conditions.

Causes of negativity of the intrapleural pressure

- Explain?

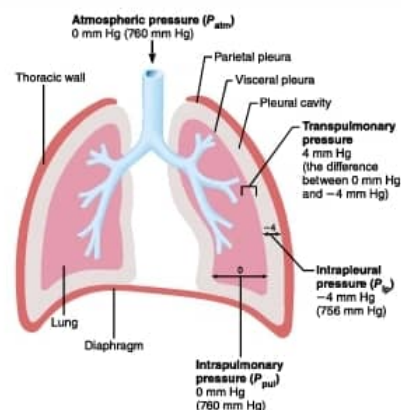
- The negativity is due to the continuous tendency of the lungs to **recoil inwards** against continuous tendency of the chest wall to expand outwards.



Negativity of the intra pleural pressure due to continuous recoil tendency of lung and expansion tendency of chest wall

4. TRANSPULMONARY (TRANAMURAL) PRESSURE (PTM)

- **Definition:** It is the difference in pressure between the intra-alveolar & intrapleural pressure.
- **Thus, transpulmonary pressure (PTM) = $P_{alv} - P_{ip}$**
- **PTM at the end of normal expiration = $0 - (-3) = 3 \text{ cmH}_2\text{O}$.**
- **PTM at the end of normal inspiration = $0 - (-6) = 6 \text{ cmH}_2\text{O}$.**
- This transpulmonary pressure is the force acting to expand the lungs.
- It is opposed by the elastic recoil of the partially expanded and, so, partially stretched lungs.
- See the diagram showing intrapulmonary and intrapleural pressure relationships



PNEUMOTHORAX

Definition

- It is the presence of air in the intrapleural space.

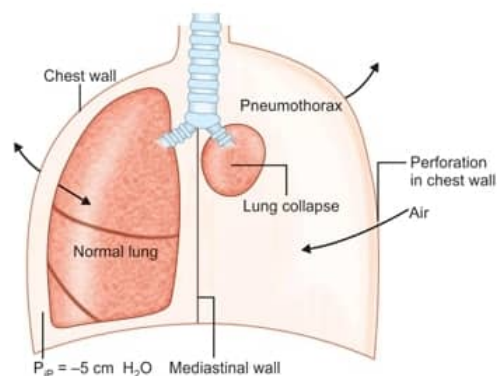
Types of Pneumothorax

a) External or opened pneumothorax:

- **Cause:** injury (stab or gun shot wound) of the thoracic chest wall.
- Air will enter from outside (atmosphere) and fill the intrapleural space.
- The negativity is lost, and the lung **collapses**.

b) Internal or closed pneumothorax:

- Although the chest wall is closed, air from the inside of the lung is found in the intrapleural space.
- **Cause:** Rupture of a surface alveolus into the intrapleural space that may take place during severe cough, as in case of **lung tuberculosis (TB)**

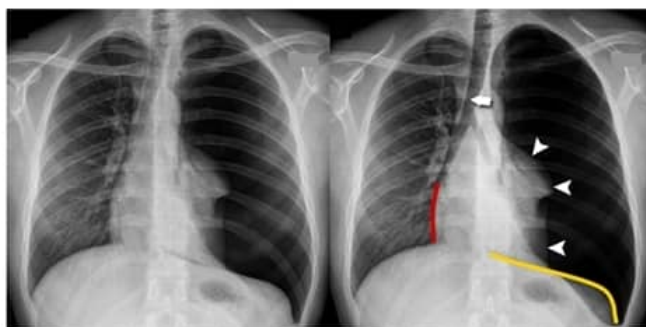
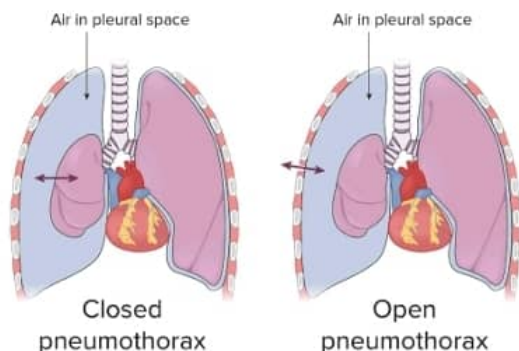


Mechanism of pneumothorax



Effects of Pneumothorax

- 1) **On the affected side:** The lung **recoils and collapses**.
 - The chest wall recoils and **expands** to reach their relaxation volumes.
- 2) **The mediastinum and the trachea** are shifted towards the **healthy** side.



BIOCHEMISTRY SECTION



SECTION IV





BLOOD BUFFERS AND ACID-BASE IMBALANCE

pH OF WATER

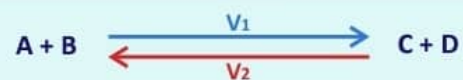
DISSOCIATION AND PH OF WATER

Law of mass action:

- The rate of a chemical reaction, at a given temperature and set conditions is **directly proportional** to the product of the molecular concentrations of the reacting substances.

- Thus for a reversible reaction that proceeds:

- 1) From left to right at a velocity = V_1
- 2) From right to left at a velocity = V_2



- V_1 is proportional to the molecular concentration of **A and B** ($K_1 = \text{Constant}$). $V_1 = [A] [B] K_1$
- V_2 is proportional to the molecular concentration of **C and D** ($K_2 = \text{Constant}$). $V_2 = [C] [D] K_2$

- At Equilibrium: $V_1 = V_2 \rightarrow [A] [B] K_1 = [C] [D] K_2$

$$\frac{V_1}{V_2} = \frac{[C] [D]}{[A] [B]} = K \text{ (equilibrium constant)}$$

- The equilibrium constant of the reaction represents the ratio between the molecular concentration of the reaction products and reacting substances.

- Pure water is a typical neutral solution, as:

- It contains equal amounts of H^+ and OH^-
- And the dissociation of H_2O is reversible.

$$K = \frac{[H^+] [OH^-]}{[H_2O]} \quad \text{OR} \quad [H^+] [OH^-] = [H_2O] K = K_w$$

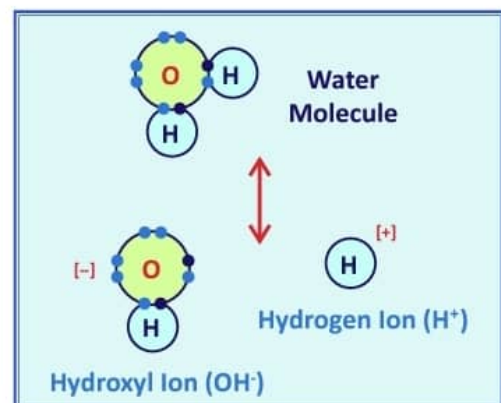
- Since the dissociation of the water is very weak, we can consider that concentration of H_2O is constant.

- K_w is known as the ionic product constant of water.

By measuring the hydrogen ion concentration of H_2O , it was found that:

$$[H^+] = [OH^-] = 1 \times 10^{-7} \rightarrow K_w = [H^+] [OH^-] = (1 \times 10^{-7}) \times (1 \times 10^{-7}) = 1 \times 10^{-14}$$

- So the ionic product of water is constant.



- This equation is applied to all water solutions, so:

- 1) Any substance that **increases** the $[H^+]$ produces a **decrease** in $[OH^-]$ & vice versa.
- 2) The product of the concentration of both at any time = 10^{-14}

- The pH: is the negative value of log of $[H^+]$ or $pH = -\log [H^+]$

- In case of water $[H^+] = 10^{-7} \rightarrow pH = -\log 10^{-7} = 7$

- The pOH: is the negative value of log of $[OH^-]$ or $pOH = -\log [OH^-]$

- In case of water $[OH^-] = 10^{-7} \rightarrow pOH = -\log 10^{-7} = 7$

So The $pH + pOH = 14$ for any substance dissolved in water

ACIDS AND BASES

1) Acids: They are substances when dissolved in H_2O produce **increase in $[H^+]$** (or termed **proton donors**) e.g. HCl.



2) Basic or alkaline solutions: They are solutions that contain **OH^- in excess of H^+** .

- In some cases by **providing OH^-** as in NaOH, and in other cases by **combining with H^+** as in amines (act as **proton acceptor**).



- **Strong Acids and Bases:**

- They **strongly dissociate** in H_2O with **marked** increase in $[H^+]$ & $[OH^-]$ respectively e.g. HCl and NaOH (their dissociation is actually **irreversible**).

- **Weak Acids and Bases:**

- They **weakly dissociate** in H_2O with **minimal** increase in $[H^+]$ & $[OH^-]$ respectively e.g. acetic acid (CH_3COOH) and NH_4OH (their dissociation is mainly **reversible**).



- **Solutions in water are classified into 3 types according to their pH:**

- 1) **Neutral solutions ($pH = 7$):** contain $[H^+] = [OH^-]$ e.g. pure water and NaCl in H_2O .
- 2) **Acidic solutions ($pH < 7$):** contain $[H^+] > [OH^-]$ or their $[H^+] > 10^{-7}$
- 3) **Basic solutions ($pH > 7$):** contain $[H^+] < [OH^-]$ or their $[H^+] < 10^{-7}$.

pH Scale	Acids							Neutral	Bases						
pH	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
pOH	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
← Increasing $[H^+]$															
Increasing $[OH^-]$ →															

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PATHOLOGY SECTION



SECTION V





NOSE, PHARYNX AND PARANASAL SINUSES

INFLAMMATORY DISORDERS

I) ACUTE INFECTIONS OF THE URT

1) Rhinitis (common cold)

- Most frequently manifesting as **common cold**.
- Manifestations:
 - Nasal congestion.
 - Watery discharge (Runny nose).
 - Slight fever.
- Pathogens:
 - Rhinovirus.
 - Para-influenza and influenza virus.
 - Respiratory syncytial virus.
- Morphology: It's an acute catarrhal inflammation with excess mucus.



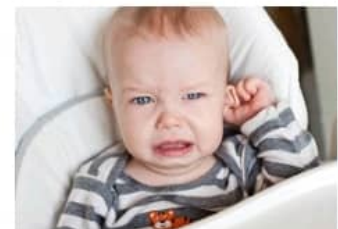
2) Acute Pharyngitis

- Manifestations:
 - a. **Mild pharyngitis** as sore throat accompany common cold.
 - b. **More severe form** is associated with marked hyperemia & exudate occurs with β -hemolytic streptococcal & adenovirus infection.



3) Otitis Media

- Definition: Infection of the middle ear, often associated with URTI.
- Clinical Picture:
 - a. **The Eustachean tube may become swollen and blocked**, leading to trapping of exudate in the middle ear.
 - b. **Eardrum perforation** may occur, following by draining of exudate.
 - c. **Chronic infection with hear loss** if inflammation persists.
- Complications: Mastoiditis, meningitis & brain abscess.



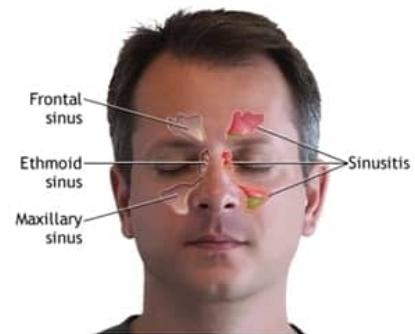
4) Acute Sinusitis

- Cause:

- Commonly preceded by **acute or chronic rhinitis**: edema impairs sinus drainage leading to retention of secretions & secondary infection
- Maxillary sinusitis can occur by **extension of a periapical tooth infection**.

- Etiology: The same pathogens as rhinitis.

- Morphology: It starts as a **catarrhal inflammation** and with secondary infection, it becomes **purulent inflammation**.



5) Acute Tonsillitis

- Infection with β -hemolytic streptococcal is common in children.

- Gross Picture: Tonsils are swollen, red with spots of pus on surface (Pus in tonsillar crypts draining suppurating lymphoid follicles).

- Microscopic Picture:

- Acute Inflammation features.
- Pus in lymphoid follicles.

- Complications:

1. Spread of infection:

- Direct:** Peritonsillar abscess (Quinzy), otitis media, pharyngitis and laryngitis.
- Lymphatic spread:** To cervical LNs causing lymphadenitis.
- Blood spread:** Bacteraemia, septicaemia, toxæmia & pyaemia.

2. Hypersensitivity to streptococcal sore throat or URT infections may result in **rheumatic fever** or **post-streptococcal glomerulonephritis**.

3. Chronicity: Chronic tonsillitis.



Acute tonsillitis and quinzy

6) Diphtheria

- Definition: It is a life-threatening disease characterized by a **pseudo-membranous inflammation** of throat & tonsils of young children.

- Etiology: Droplet infection by *Corynebacterium diphtheria*.

- Incubation period: 2-5 days.

- Sites: Tonsils, soft palate, larynx and trachea.

- Pathological features:

- Pseudo-membranous inflammation:** See next page.
- Acute cervical lymphadenitis:** Cervical lymph nodes on both sides are enlarged, separate, mobile, soft, painful and tender.
- Severe acute toxæmia.**

- Complications:

- Asphyxia:** Due to oedema, membrane and laryngeal spasm.
- Acute toxæmia** may cause:
 - Toxic myocarditis and acute heart failure.
 - Cloudy swelling and focal necrosis of parenchymal organs.
 - Haemorrhage and necrosis of supra-renal cortex resulting in acute insufficiency.
 - Temporary paralysis of muscles of palate, pharynx, larynx, eye, face, etc.. (nerve affection by bacterial toxins).



Diphtheria



Pseudo-membranous inflammation

- Gross Picture:

- **Multiple small yellowish foci of necrosis** gradually enlarge fuse together and form a continuous membrane.
- The membrane is slightly raised and adherent, but if removed by force it leaves **bleeding ulcerating surface** and re-forms again.

- Microscopic Picture:

- **The membrane is formed of:** fibrin network entangling dead epithelial cells, polymorphs, pus cells and macrophages.
- **The submucosa** shows dilated congested vessels, oedema, fibrin and acute inflammatory cells.



II) ALLERGIC DISEASES OF THE URT

1) Allergic Rhinitis

- **Definition:** Atopic disease, immunoglobulin E (IgE)-mediated immune reaction caused by different antigens.

- **Etiology:** Allergens as Pollen, grass, dust ...etc produce **Type I hypersensitivity (atopy)**.

- **Gross Picture:** Marked mucosal edema and erythema of the nasal mucosal membranes

- **Microscopic Picture:** Acute allergic inflammation rich in **eosinophils** and increased submucosal edema.

- **Complications:** Repeated attacks result in nasal polyp rich in plasma cells, lymphocytes and eosinophils.



2) Nasal Polyp

- **Definition:** Edematous mucosal protrusions infiltrated by eosinophils and plasma cells.

- **Cause:** Nasal polyps occur with recurrent allergic rhinitis.

- **Gross Picture:** Multiple, soft, pink polyps projecting from mucosa of nose and sinuses

- **Microscopic Picture:**

- **Epithelium:** Pseudostratified columnar ciliated (respiratory), but with chronic irritation it may changes to squamous epithelium (**metaplasia**)
- **Basement membrane:** thickened and pink.
- Thick, very wide edematous core with many eosinophils, lymphocytes and plasma cells.
- **Mucous glands:** increase in number (**hyperplasia**).

- **Complications:** when multiple or large, they obstruct the airway

- **Nasal obstruction** leading to 2^{ry} infection, spread of the infection & epistaxis.



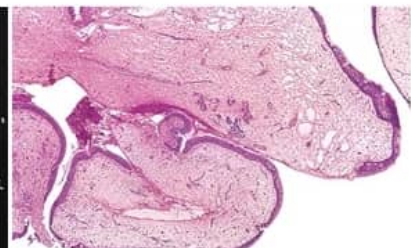
Nasal polyps: Grossly,



Endoscopy



CT image



Microscopic Picture

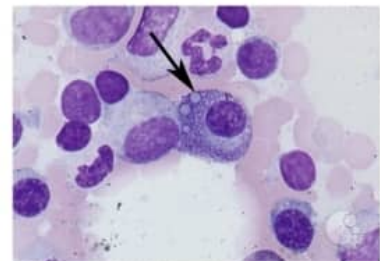
III) HYPERPLASTIC DISEASES (ADENOIDS)

- **Definition:** Hyperplastic lymphoid tissue at the **posterior wall** of the nasopharynx, due to chronic infection in infants and children.
- **Gross Picture:** Mouth breathing with characteristic adenoid face:
 - Opened mouth.
 - Short upper lip.
 - Protruding upper incisors.
 - Narrow nasal openings.
 - Absent nasolabial folds.
- **Complications:** spread of infection leading to otitis media & pharyngitis.

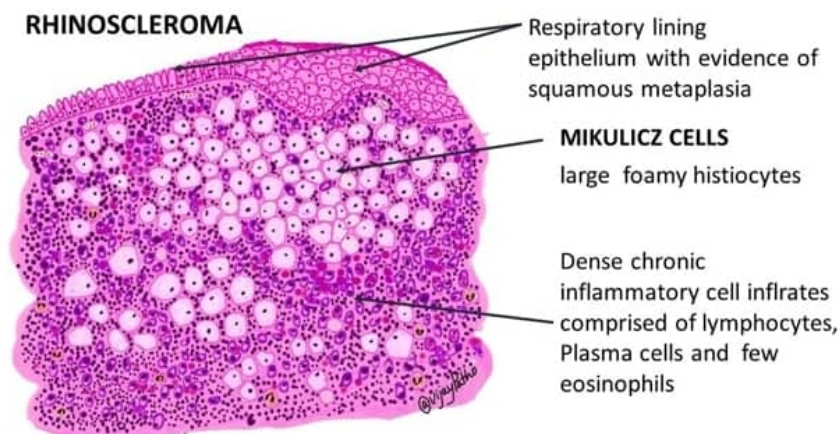


IV) CHRONIC SPECIFIC INFECTIONS (RHINOSCLEROMA)

- **Definition:** A granulomatous inflammation of the nose.
 - May affect other site as the larynx (laryngoscleroma) and pharynx (Pharyngoscleroma).
- **Cause:** Due to infection with **Klebsiella rhinoscleromatis** bacteria which is endemic in Egypt.
- **Pathogenesis:** The bacteria multiplies in macrophages producing a foamy appearance of the cytoplasm.
 - These cells are called **Mickulicz cells**.
- **Gross Picture:** The lesions appear as small hard nodular mass or masses filling the nasal cavity.
- **Microscopic Picture:** There is a chronic inflammatory infiltrate involving the submucosa. The inflammatory cells comprise:
 1. **Mickulicz cells:**
 - These are the predominant cells.
 - They are macrophages with foamy cytoplasm.
 - They are large with a small pyknotic nucleus.
 2. **Plasma cells and Russell bodies** (Plasma cells with hyaline change due to accumulated immunoglobulin)
 3. Later there is **fibrosis**.
- **Complications:**
 - Nasal obstruction and deformity.
 - Ulceration and secondary infection.



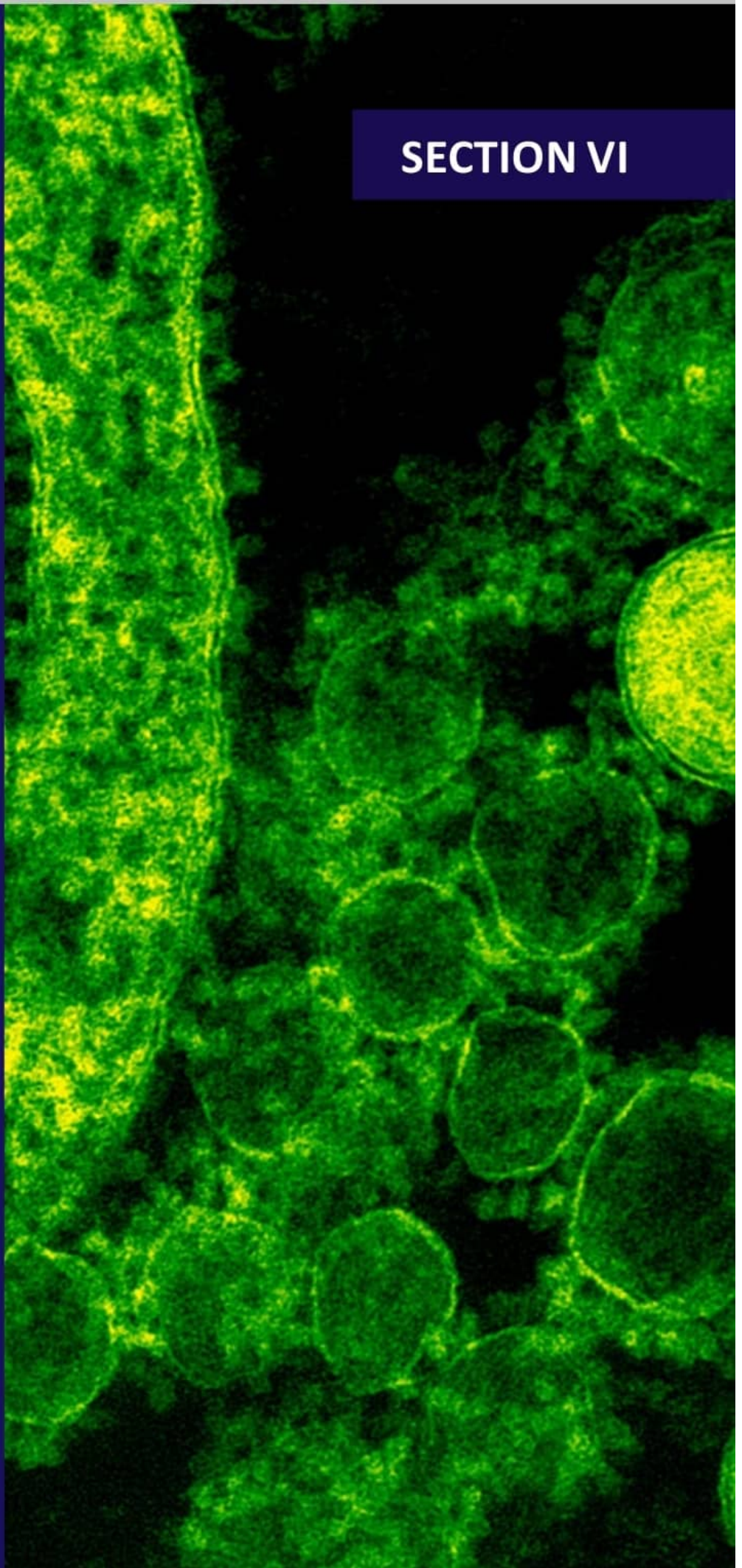
Russell bodies





MICROBIOLOGY SECTION

SECTION VI



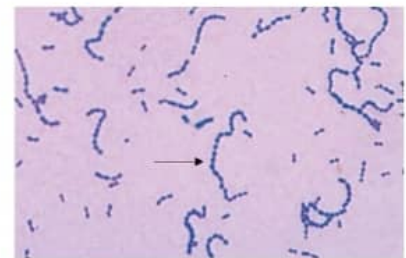


BACTERIAL INFECTIONS OF RESPIRATORY TRACT

STREPTOCOCCUS

Morphology

- Gram-positive ovoid cocci arranged in **chains** or **pairs**.
- **Non**-motile and **Non**-spore forming.
- **Catalase negative**: **The key test** for discriminating streptococci from the catalase-positive staphylococci.
- **Culture**: requires **enriched** media containing blood or serum.
- **The most important species**:
 1. Streptococcus pyogenes
 2. Streptococcus agalactiae
 3. Streptococcus pneumoniae



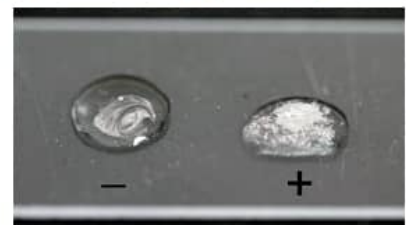
Classification

I) According to Haemolysis on blood agar:

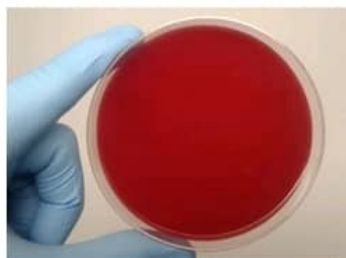
- 1) **β -hemolytic Strept**: **Complete** hemolysis, e.g.: Strept. pyogenes.
- 2) **α -hemolytic Strept**: **Partial** hemolysis, e.g.: Strept. viridans & Strept. pneumonia.
- 3) **γ -hemolytic Strept**: **No** hemolysis.

III) Lancefield classification: Streptococci are classified according to a **group-specific carbohydrate antigen** into groups A to U.

- **Antibodies** against these group antigens are used for identification of streptococcal species.



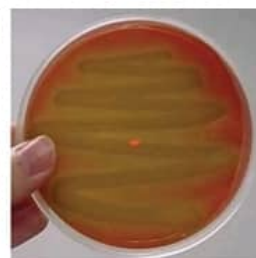
Catalase test



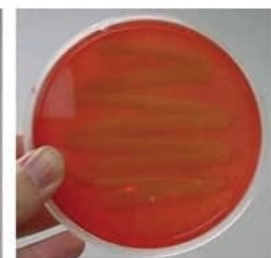
Blood agar



β -hemolysis



α -hemolysis



γ -hemolysis

STREPTOCOCCUS PYOGENES INFECTIONS

I) Localized infections	II) Invasive infections
1. Pharyngitis (sore throat and tonsillitis). 2. Scarlet fever. 3. Skin and soft tissue infections.	1. Puerperal fever. 2. Acute endocarditis. 3. Necrotizing fasciitis. 4. Toxic shock syndrome (TSS).

I) Localized Infections

1. Pharyngitis (sore throat and tonsillitis)

- It is **the commonest infection** caused by *S. pyogenes*.

- Symptoms:

- Pain. • Redness. • Fever.
- Swelling of posterior pharynx.
- Grayish white tonsillar exudate

- Transmission: is usually transmitted via respiratory droplets.



2. Scarlet fever

- **Cause:** It occurs when the streptococcal infection (especially pharyngitis) is caused by an **erythrogenic toxin-producing *S. pyogenes* (Erythrogenic toxin A)**.

- **Manifestations:**

- Sore throat.
- Pharyngitis.
- Scarlet red rash (sandpaper rash).
- Strawberry tongue.



Pharyngitis



Scarlet rash



Strawberry tongue

3. Skin and soft tissue infections

A) Impetigo (pyoderma): It is a local infection of the **superficial** layers of the skin with blisters and denuded surface covered with crusts.

B) Cellulitis: It is infection of the **deep** layers of the skin.

C) Erysipelas: It is a form of cellulitis accompanied by **fever** and **systemic toxicity**.



Impetigo



Erysipelas



Cellulitis

II) Invasive Infections

1) Puerperal Fever

- This is a **life-threatening** infection of the **endometrium** and surrounding structures complicating delivery or abortion.
- Septicaemia and toxic shock may occur.

2) Acute Endocarditis

- This **fatal** condition can occur in individuals with **normal or damaged** heart valves.

3) Necrotizing Fasciitis

- **Cause:** associated with **severe tissue destruction** particularly associated with **SPE-B (protease)**.
- The destructive nature of this condition led to the term "**flesh-eating bacteria**".

4) Toxic Shock Syndrome (TSS)

- **Cause:** This condition is mediated by the production of **SPE- A, B & C**.
- **Manifestations:** It often begins with **skin wounds or minor traumas** and rapidly deteriorates leading to **necrotizing soft tissue infections**.
- **Complications:** Shock, renal failure and acute respiratory distress syndrome.



Acute Endocarditis



Necrotizing Fasciitis



Soft tissue infections in TSS

Laboratory Diagnosis

A. Specimens

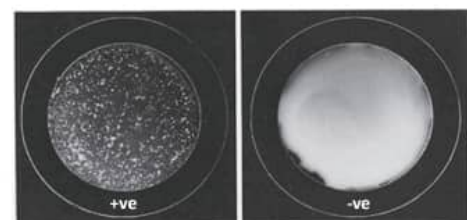
- **Include:** throat swab, pus, blood (in invasive infections)...etc.



Throat swab

B. Direct detection in clinical specimens

1. **Gram-stained smears** are useful **only** in cases of skin and soft tissue infections, **Why?**
 - As *S. pyogenes* **cannot be distinguished from the normal oral streptococcal flora**.
2. **Detection of Lancefield group A streptococcal antigen** in throat swab by **latex agglutination test** using group A antibody.



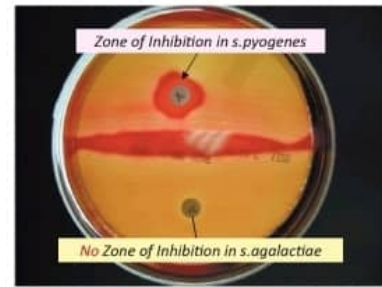
Latex agglutination test

C. Cultivation

- Specimens other than blood should be **plated directly onto blood agar** and incubated at 37°C.
- Blood samples should be cultivated by **the blood culture technique**.
 - Subcultures are plated on blood agar.

D. Identification

- **After 24h incubation**, the growth should be examined for colony morphology, Gram stain and catalase production:
 1. **On blood agar**: Colonies are surrounded by beta-haemolysis.
 2. **Gram-stained film**: Gram positive cocci in chains.
 3. **Catalase test**: negative.
- *S. pyogenes* can further be identified by:
 1. **Growth inhibition by bacitracin** (i.e. it is bacitracin-sensitive).
 2. **Agglutination by group A antibody**.



Bacitracin Test

POST-STREPTOCOCCAL SEQUELAE

(Acute rheumatic fever & Acute Glomerulonephritis)

- **Definition**: These are non-suppurative inflammatory condition which occurs as a result of **immunologic response** to streptococcal antigens.
- **Pathogenesis**:
 - They occur weeks following a **local** infection with *S. pyogenes*.
 - The organisms affect an organ that was **not** infected by the streptococci.
- **Serotypes**: Some streptococcal M serotypes are:
 1. **Rheumatogenic** and cause acute rheumatic fever (ARF)
 2. **Nephritogenic** and cause acute glomerulonephritis (AGN).



1) Acute Rheumatic Fever (ARF)

- **Incidence**: ARF may affect any age (but mostly 5-15 years).
- **Precipitating Infection**: It develops **2-3 weeks** following streptococcal infections of the upper respiratory tract, e.g. sore throat or acute follicular tonsillitis.
- **Rheumatogenic strains**: Include serotypes 1, 3, 6, 14, 18, 19, 24, 27 and 29.
- **Cause**: It is due to the formation of **antibodies** to streptococcal M protein, which **cross-react** with antigens of joints, heart and brain tissue (**autoimmune reaction**).
- **Manifestations**:
 - Fever and Malaise.
 - Carditis.
 - Migratory polyarthrititis.
 - Erythema marginatum.
 - Chorea.
 - Subcutaneous nodules.
- **Recurrence**: The immunological process is **exacerbated** by recurrence of streptococcal pharyngitis leading to **valvular damage**.
- **Diagnosis**: is usually based on the **modified Jones criteria** (See next page).
- **Prevention**: Preventable if the patient is treated within the **first 10 days** of acute pharyngitis.
 - **Long-term antibiotic prophylaxis** using penicillin is recommended following a single attack.



Erythema marginatum



Migratory polyarthrititis



Chorea



Subcutaneous nodules



Modified Jones Criteria

- It is an approach for ARF diagnosis that include major and minor criteria.
- Diagnosis requires an **evidence** of recent *S. pyogenes* infection together with:
 - **Two** major criteria or • **One** major and **two** minor criteria.

- Evidence of Recent Streptococcal Infection:

1. A history of acute tonsillitis (or scarlet fever).
2. Positive throat swab culture for *S. pyogenes*.
3. Elevation of antistreptolysin O (ASO) titre above 200 units.
 - A titre of up to 200 units is considered normal.



- Major Criteria: **CANCER**:

1. **C**arditis.
2. Migratory poly-**A**rthritis.
3. Subcutaneous **N**odules.
4. **C**horea.
5. **E**rythema marginatum.

- Minor Criteria: **PEACEFL**

1. **P**rior history of RF.
2. **E**CG with PR prolongation
3. **A**rthralgia
4. **C**RP increased
5. **E**levated ESR
6. **F**ever.
7. **L**eukocytosis

2) Acute glomerulonephritis (AGN)

- **Incidence:** AGN may affects children more than adults.
- **Precipitating Infection:** It develops **one week** following **skin** infections, e.g. impetigo.
- **Nephritogenic strains:** Include serotypes 2, 4, 12 and 49.
- **Cause:** It is due to **deposition of antigen-antibody complexes on the glomerular basement membrane**, provoking an inflammatory response that damages kidney (**type III hypersensitivity**).
- **Manifestations:**
 - Edema.
 - Nitrogen retention.
 - Hypertension
 - Presence of blood, albumin and granular casts in urine (**smoky urine**).
- **Fate:** Most **recover completely** and chronic renal failure rarely occurs.
- **Diagnosis:** There are **elevated antibody titres for anti-DNase B**.
 - **The Antistreptolysin O titer (ASO):**
 - a. In glomerulonephritis following pyoderma or skin infection: generally **low**.
 - b. After throat infection (rare): may be **higher**.
- **Recurrence:** Reinfection with streptococci **rarely** leads to recurrence of AGN
 - Antibiotic prophylaxis is unnecessary.
 - Treatment of the streptococcal skin disease or pharyngitis **doesn't prevent AGN**.



Smoky (bloody) urine



Oedema of the face



Treatment and Prevention of *S. pyogenes*

1. **Penicillin** is still uniformly effective in treatment of *S. pyogenes* infections.
 - **Long acting penicillin** is used as a chemoprophylaxis in recurrent *S. pyogenes* infections to prevent **repeated rheumatic attacks**.
2. **Erythromycin** is used as an alternative in penicillin-allergy.

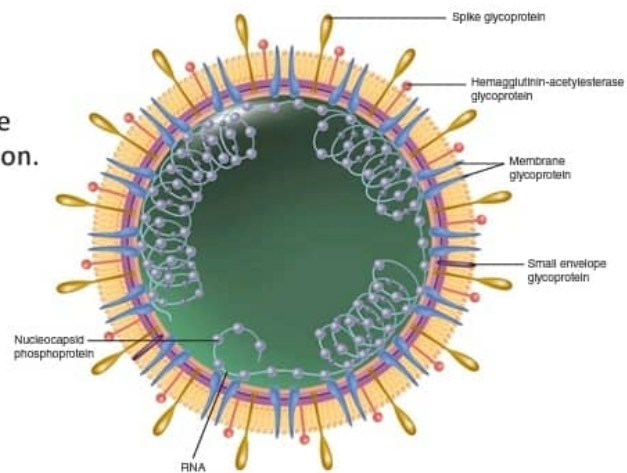
CORONAVIRUSES

- Morphology:

- **Enveloped ssRNA viruses**
- The family received the name "corona" because of **the crown-like appearance** of the surface projections which surround each virion.

- Diseases:

1. Common cold
2. Severe acute respiratory syndrome (SARS)
3. Middle east respiratory syndrome (MERS)
4. Coronavirus disease 2019 (Covid-19)



1. Common cold

- It is a mild upper respiratory tract illness
- Coronaviruses are **the second most common cause** of common cold following the rhinoviruses.

2. Severe acute respiratory syndrome (SARS)

- **Caused by:** the SARS coronavirus (SARS-CoV)

- Outbreak:

- SARS-CoV was first detected in China in 2002 and subsequently spread to many countries.
- This outbreak subsided, and no new cases have been identified since 2004.

- Clinical picture:

- It is a **severe atypical pneumonia** characterized by fever, dry cough and dyspnea.
- **Diarrhoea** has been a prominent feature of early illness in some cases.

- **The incubation period:** ranges from 2-10 days.

- **The mortality rate:** is about 10%.

3. Middle east respiratory syndrome (MERS)

- **Caused by:** It is a severe acute respiratory illness caused by the recently identified MERS coronavirus (MERS-CoV).

- **Outbreak:** MERS-CoV was first reported in Saudi Arabia in 2012 and since then, it has been reported in other countries mainly in the Middle East.

- **The mortality rate:** is about 35%.



Diagnosis

- **Common cold** is primarily a clinical one.
- **SARS and MERS** are only suspected if there is prior history of travel to an endemic area.
- **Laboratory confirmation is done by detection of antibodies and by PCR.**

4. CORONAVIRUS DISEASE 2019 (COVID-19)

- Definition: Coronavirus disease 2019 (COVID-19) is an infectious disease caused by **severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)**.

- Beginning:

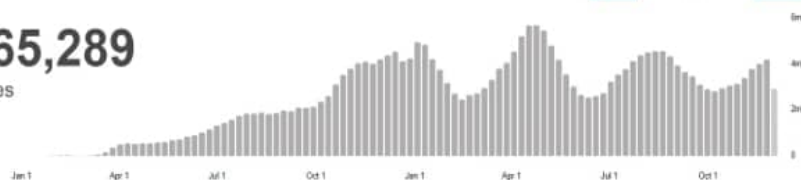
- COVID-19 was first identified in December 2019 in Wuhan, Hubei, China and has resulted in an **ongoing pandemic**.
- By October 2020, approximately, 38 millions of cases been reported across 188 countries, but the WHO estimates that around 800 million people in total may have been infected.
- By December 13w,2021, Live updates WHO confirmed **270,510,989 Cases**, **5,324,329 Deaths** and **243,304,544 Recovered**.

- Severity: The COVID-19 disease has killed approximately 1 million people (overall mortality rate is 2.3%) and more than 26 million have recovered.

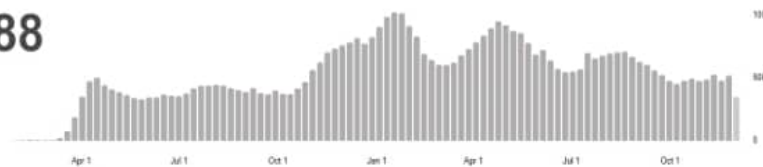
Globally, as of 4:48pm CET, 10 December 2021, there have been **267,865,289 confirmed cases** of COVID-19, including **5,285,888 deaths**, reported to WHO. As of **9 December 2021**, a total of **8,158,815,265 vaccine doses** have been administered.

Global Situation

267,865,289
confirmed cases



5,285,888
deaths



Source: World Health Organization
Data may be incomplete for the current day or week.

Virology

- **SARS-CoV2** is closely related to the original SARS-CoV.
- It is thought to have an **animal (zoonotic) origin**.
- It is **96% identical at the whole genome level** to other bat coronaviruses.



COVID-19 Variants

- Delta Variant:

- Reported to WHO in July 2021, first detected in India.
- **Causes more infections and spreads faster** than early forms of SARS-CoV-2

- Omicron Variant:

- Reported to WHO in November 2021, first detected in Botswana and South Africa.
- WHO named the **B.1.1.529 Omicron** & classified it as a **Variant of Concern (VOC)**.



OMICRON



DELTA

FIRST DETECTION:	NOVEMBER 2021	July 2021
COUNTRY:	SOUTH AFRICA	INDIA
PANGO LINEAGE:	B.1.1.529	B.1.617.2
REINFECTION RISK:	HIGH	HIGH

Pathogenesis

Mode of transmission

- **The principal mode of transmission:** respiratory droplets carrying infectious virus (generally within a space of one metre) in confined spaces, hence the importance of social distancing and masks in its prevention.
- **Additional methods:** include:
 - a. Contact transmission (e.g., shaking hands)
 - b. Contaminated surfaces.



Important Note

- How do large respiratory droplets (60-100 μm) travel through air?

- More than 6 meters away at a velocity of 50 meters/second (m/s) during sneezing
- More than 2 meters away at a velocity of 10 m/s during coughing
- Less than 1 meter away at a velocity of 1 m/s during breathing

Period of infectivity of SARS-CoV2

- Spread from:

- a. Infected person for up to two days prior to symptoms' onset
- b. Asymptomatic people.

- Period of Infectivity: for 7-12 days in moderate cases and up to two weeks in severe cases.

- Asymptomatic persons: account for approximately 40-45% of SARS-CoV2 infection.

- These asymptomatic carriers can transmit infection to others for an extended period, perhaps more than 14 days, and therefore, contribute to substantial spread of the disease.

Incubation period 2-14 days.

Symptoms

- Fever or chills
- Fatigue
- Muscle or body aches
- New loss of taste or smell
- Congestion or runny nose
- Cough
- Shortness of breath or difficulty breathing
- Headache
- Sore throat
- Nausea, vomiting & diarrhea



Fever



Chills



Cough



Muscle aches



Congestion



Loss of smell



Sore Throat



Nausea & Vomiting

Prevention and treatment

- Recommended measures to prevent COVID-19 infection: include:

- **Wearing a mask** in public places
- **Staying at least two meters away** from other people
- **Washing hands** frequently
- **Avoiding crowds** and confined spaces
- **Respiratory hygiene/cough etiquette:**
 - a. Covering mouth and nose when coughing or sneezing
 - b. Washing hands or antiseptic hand hygiene every time the mouth or nose is touched
- **Quarantine** • **COVID-19 vaccines.**

- Management:

- There is **No specific treatment** for COVID-19
- It involves the treatment of symptoms, supportive care and isolation measures.



COVID-19 Vaccines

- They are now widely available for people ages 5 years and older.
- All currently approved or authorized COVID-19 vaccines probably reduce the risk of severe illness

company	Type	Doses	Efficacy	Storage
Pfizer-BioNTech	mRNA	Two	95%	-70 °C
Moderna	mRNA	Two	94.5%	-20 °C
Sputnik V	Viral vector	Two	91.6%	Regular Fridge
Novavax	Adjuvanted recombinant	Two	89.3%	Regular Fridge
Sinopharm	Inactivated	Two	86%	Regular Fridge
AstraZeneca	Viral vector	Two	76% - 82%	Regular Fridge
Janssen (J&J)	Viral vector	One	66% - 72%	Regular Fridge

BIOLOGICAL WARFARE AND BIOTERRORISM

- **Definition:** it is the intentional use of biological agents that cause disease or death in people, animals or plants.

- **Forms:** They may be:

- **Naturally occurring**
- **Genetically modified** form to increase their virulence or resistance to drugs or vaccines.

- **Classification:** They are classified into categories, based on:

- **The ease of dissemination**
- **Their ability to cause morbidity and mortality.**

- **Agents in the highest risk category:** These agents may be bacteria, viruses or toxins:

- | | | |
|-----------|--|------------|
| • Anthrax | • Smallpox | • Botulism |
| • Plague | • Viral haemorrhagic fever (e.g., Ebola virus) | |



PHARMACOLOGY

SECTION



SECTION VIII



B) METHYLXANTHINES DRUGS

- Theophylline is **the most commonly used preparation in Egypt**.

Pharmacokinetics

- Preparations:

- **Theophylline salts** different formulations are used to improve absorption .
- **Sustained release tablets** are used to achieve serum levels for **12 h** for less frequent administration, less fluctuation in serum levels, more effective control for nocturnal asthma.

- **Therapeutic plasma concentration:** between 5-20 mg/L.

- Toxic levels:

- Higher than 20 mg/L:** Anorexia, nausea, vomiting, headache, anxiety and abdominal discomfort.
 - Sometimes these symptoms **start at 15 mg/L**.
- Higher levels than 40 mg/L:** arrhythmia or seizures.

- **Metabolism:** by liver.

- Hepatic microsomal **inhibitors:** increase plasma theophylline levels → **Toxicity**.
- Hepatic microsomal **inducers:** decrease plasma theophylline levels → **Sub-therapeutic doses**.



- **Measuring blood levels for patients** on theophylline is important as it has **narrow therapeutic window**.

- **Therapeutic and toxic effects** are related to plasma concentration.

Mechanism of action

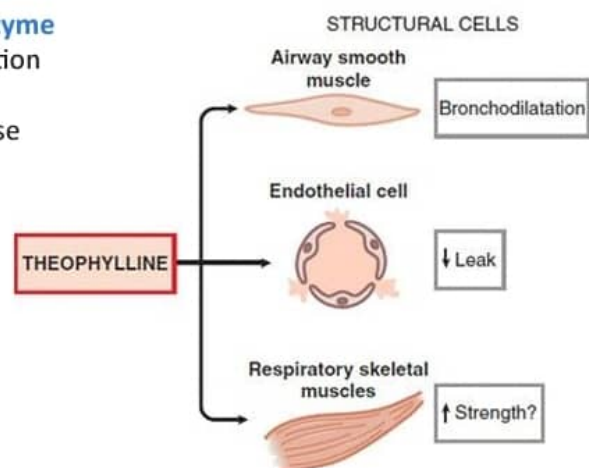
1. **Methylxanthines inhibit phosphodiesterase enzyme type 4 (PDE4)** that result in increased accumulation of cellular cAMP and cGMP.

- This causes **direct bronchodilatation** & decrease some inflammatory reactions in the airways.
- The inhibition of PDE4 in inflammatory cells **reduces the release of cytokines** and results in decreasing cell migration.

2. **Block adenosine receptors** leading to:

- Bronchodilatation
- Decrease histamine release.

3. **Improve diaphragmatic contractility.**



Therapeutic uses

- Theophylline is **the most effective xanthine as bronchodilator**:

1. **Aminophylline slowly IV** relieves acute attacks of bronchial asthma and used in quick relief medications (**asthma reliever**).
2. **Sustained release (SR) theophylline** reduces the severity and number of exacerbations in chronic asthma and used in long term control medications (**asthma controller**).

Adverse Effects of Theophylline

GIT	Neurological		CVS	Allergic
- Abdominal pain - Anorexia - Diarrhoea - Nausea & Vomiting	- Confusion - Headache - Major seizures - Tremor	- Dizziness - Irritability - Insomnia - Tinnitus	- Cardiac dysrhythmias - Flushing	- Rashes - Pruritis - Fever
				
Tremors	Nausea & vomiting	Abdominal pain	Anorexia	
				
Tinnitus	Insomnia	Dizziness	Rashes	

C) ANTI-MUSCARINIC DRUGS

- **Members:** Antiasthmatic atropine substitutes Includes:

1. Ipratropium.
2. Tiotropium.

- **Route of Administration:** are used & given by inhalation.



Mechanism of Action

- **Block muscarinic receptors** in the airways, so they:
 - a. Prevent bronchospasm.
 - b. Decrease mucus gland hypersecretion seen in asthma.

Clinical Uses

- They are effective bronchodilators.
- Anti-muscarinic drugs are used in:
 1. Patients intolerant to inhaled β agonists.
 2. In addition to inhaled β agonists to add to their bronchodilator action.
 3. Drugs of choice in bronchospasm resulting from non selective β blockers.
 4. Patients with chronic obstructive pulmonary disease (COPD).

- **The degree of involvement of parasympathetic pathway** in bronchospasm differs from one subject to another.
- In some patients, these drugs are very effective in others, only modestly.



Advantages Over Atropine

- 1) More selective bronchodilator action.
- 2) Decrease the augmentation of mucus secretion which occurs in bronchial asthma.
- 3) Increase the mucociliary clearance of secretion.

TREATMENT OF DRY NON-PRODUCTIVE COUGH

ANTI-TUSSIVE DRUGS

- Anti-tussive Drugs which suppress cough are used and classified into:

A. Peripheral Anti-tussives: Suppress sensory nerve endings which initiate cough reflex.

B. Central Anti-tussives: Suppress cough center in the medulla.

A. Peripheral Anti-tussives

- Anti-tussives acting peripherally:

1. **Demulcents:** in pharyngitis and sore throat e.g. liquorice (pastilles-lozenges syrup).

2. **Steam inhalation:** in tracheo-bronchitis e.g. Tincture benzoin Co or menthol in a dose 1 tea spoonful added to 500 ml of boiling water.

- **Action:** It promotes secretion of protective mucus.

3. **Local anesthetics:** (when pulmonary bed is irritated)



Steam inhalation



liquorice

B. Central Anti-tussives

- Anti-tussives acting centrally:

- **Mechanism of Action:** Direct inhibition of the cough center in medulla, so they suppress cough reflex.

- **Non-narcotic antitussives (non-addictive):**

a) **Dextromethorphan:** used in treatment of chronic non-productive cough.

b) **Narcotine.**

c) **Non-opiate synthetic derivatives:** Benzonatate, Oxeladin.



- **Benzonatate** is rarely used because it is **not** as effective as other antitussives.

• **Action:** It acts by anaesthetizing the stretch receptors in the respiratory tract, lung tissues and pleura, and thereby reducing the cough reflex.



TREATMENT OF PRODUCTIVE COUGH

EXPECTORANTS

- **Definition:** Expectorants are ingredients that **increase** airway secretions.

- **Action:** They do this by increasing the water content of the secretions which decreases their stickiness, making them easier to cough up.

- **Used in:** relieve chest congestion that occurs because of a cold, the flu, or allergies.

- **Members:**

1. **Guaifenesin:** **OTC (Over-the-counter)** expectorant used to relieve symptoms of respiratory conditions (Acute respiratory tract infections as common cold, acute bronchitis)

2. **Potassium iodide (Saline expectorants)**

